

Strategic Investors and Exchange Rate Dynamics

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April 30, 2026

Abstract

We study how the exchange rate dynamics are influenced by the presence of heterogeneous investors with varying degrees of price impact. Leveraging data from the U.S. Commodity Futures Trading Commission (CFTC) on investors' currency positions, we show that foreign exchange rate markets display a significant level of concentration. We develop a monetary model of exchange rate determination that incorporates heterogeneous investors with different degrees of price impact. Our model predicts that the presence of price impact amplifies the exchange rate's response to non-fundamental shocks while dampening its response to fundamental shocks. As a result, investors' price impact contributes to the disconnect of exchange rates from fundamentals, increases the excess volatility of exchange rates, and makes excess return more predictable. We provide empirical evidence in line with our theoretical predictions, using data on trading volume concentration from the US CFTC foreign exchange rate market for 10 currencies spanning from 2006 to 2016.

JEL Codes: F31, G11, G15

Keywords: Exchange Rate, Investors' Heterogeneity, Price Impact, Strategic Investors, Dispersed Information, Exchange Rate Puzzles, Exchange Rate Disconnect, Excess Volatility, Excess Return Predictability.

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1 Introduction

Three well-known puzzles in international economics are the limited explanatory power of macroeconomic fundamentals in accounting for exchange rate fluctuations (the exchange rate determination puzzle), the excessive volatility of exchange rates relative to fundamentals (the excess volatility puzzle), and the failure of uncovered interest parity (the UIP puzzle) (Meese and Rogoff, 1983; Obstfeld and Rogoff, 2000; Fama, 1984).¹ Recent evidence from the microstructure approach to exchange rates suggests that investor heterogeneity plays a crucial role in understanding exchange rate dynamics and determination. For example, these puzzles can be explained by the rational confusion arising from information heterogeneity (Bacchetta and Van Wincoop, 2006). Similarly, exchange rate behavior is linked to order flow, which, in turn, is associated with the heterogeneity among investors (Lyons et al., 2001; Evans and Lyons, 2006). This paper investigates how the presence of large investors who internalize the impact of their trading on currency prices can potentially influence exchange rate fluctuations.

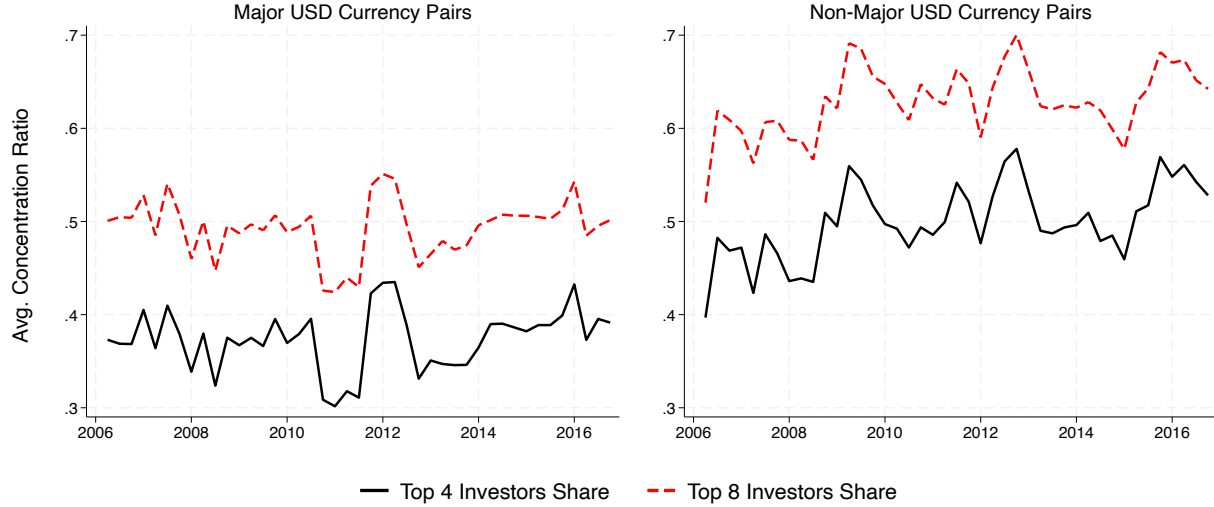
Foreign exchange markets are historically characterized by an high degree of concentration, with a relatively small number of investors holding a substantial presence in the markets through their currency positions. Figure 1 reports the average concentration ratios by currency groups (major and non-major currencies), computed as the share of net open interest positions held by the largest four and eight investors operating the principal foreign exchange markets from 2006 to 2016.² Two facts stand out. Firstly, the eight (four) largest entities collectively held approximately 50% to 70% (40% to 60%) of the open interest positions in the foreign exchange market. Secondly, non-major USD currency pairs are notably more concentrated, by about 20%, compared to major ones.³ Figure 1 suggests the potential

¹Meese and Rogoff (1983) show that macroeconomic models have lower predictive power compared to a random walk model. Similarly, Obstfeld and Rogoff (2000) show that exchange rates exhibit significantly more fluctuations than their underlying fundamentals.

²Appendix A provide more information on the data.

³Figure 3 in Appendix A shows the concentration ratios for all individual currencies from 2006 to 2016. The Brazilian Real, Russian Ruble, and New Zealand Dollar are the currency pairs with the highest concentration, while the Euro and Canadian Dollar exhibit the lowest. Figure 4 in Appendix A shows qualitatively similar patterns when concentration is measured by the number of entities trading each currency (on average, 10 to 25 entities actively trade, with more traders active in major currency markets). The high concentration measured here aligns with other pieces of evidence, such as the BIS Triennial Survey of Foreign Exchange Markets or the NY FED OTC Foreign Exchange Market Survey. However, these surveys are limited in their scope, frequency of observation, or coverage. The leading foreign exchange market survey, conducted by Euromoney and covering global markets, reveals that around 25 entities transact 70% of the total turnover. Additional information can be found

Figure 1: Market Concentration – U.S. CFTC



Notes: The figure shows the average concentration ratio of net open interest positions held by asset managers, institutional investors, and leveraged funds across currencies, divided between major and non-major currency groups. We consider the share held by the eight and the four largest entities in each market. Concentration ratios are computed on ‘Net Position’, meaning that they are calculated after offsetting each trader’s long and short positions. Major currency pairs consist of the United States Dollar paired with the Euro, British Pound, Japanese Yen, and Swiss Franc. Non-Major currency pairs include the United States Dollar paired with the Australian Dollar, Canadian Dollar, New Zealand Dollar, Mexican Peso, Brazilian Real, and Russian Ruble. The data is sourced from the U.S. Commodity Futures Trading Commission (CFTC) and spans from 2006 to 2016, with quarterly averages for each currency pair. Appendix A provides additional details regarding the data used.

for individual investors to influence currency prices (Kyle, 1985).⁴

We embed heterogeneity in price impact into a two-country, dynamic monetary model of exchange rate determination to analyze the potential effects of strategic behavior on exchange rate fluctuations. In our theoretical framework, investors face an international portfolio choice problem with noise shocks. Departing from the conventional assumption of price-taking investors, we introduce a continuum of investors with varying degrees of price impact. One group of investors is atomistic and competitive, acting as price takers, while

<https://www.euromoney.com/article/b1lp5n97k4v6j0/fx-survey-2020-press-release>.

⁴This potential is not just theoretical. The 2013 London FX fixing scandal, the 2015 Swiss franc peg removal, and the 2016 British pound flash crash all demonstrate how concentrated trading flows of large traders can significantly impact currency markets, leading to substantial price movements and market disruptions. Despite significant institutional reforms implemented in 2015, there are indications that market manipulation may not have completely ceased (Osler, 2014; Osler et al., 2016; Cochrane, 2015).

the remaining group consists of a finite number of strategic investors with non-zero market power, who behave oligopolistically and internalize the effects of their trading decisions on equilibrium prices. In equilibrium, the exchange rate is determined as a weighted average of fundamentals, such as interest rate differentials, and noise shocks, with the weights depending on market structure and the average size of strategic investors.

We present a set of theoretical predictions on how strategic investors influence exchange rate dynamics. First, the presence of strategic investors amplifies the exchange rate’s response to noise shocks while dampening its response to fundamental shocks, thereby exacerbating the excess volatility puzzle. In the model, a positive noise shock decreases the residual demand for foreign assets, driving up their price and the exchange rate, without changes in fundamentals. Conversely, a negative fundamental shock increases the excess returns on foreign assets, resulting in an appreciation of the foreign currency. Strategic investors, aware of their price impact, adjust their trading positions less aggressively, which amplifies the effect of noise shocks and weakens the impact of fundamental shocks on the exchange rate.

Our theoretical framework predicts that heterogeneity in price impact can contribute to exchange rate predictability, the exchange rate disconnect and the excess volatility of the exchange rate relative to fundamentals. First, in the model, systematic deviations from Uncovered Interest Parity (UIP) arise due to a non-zero net supply of foreign assets. Strategic behavior amplifies UIP deviations, making the Fama coefficient more negative and reflecting a stronger influence of interest rate differentials on excess returns and risk premiums. Second, the presence of strategic investors leads to a reduction in the information loading factor of the exchange rate (reduced informativeness), meaning that the exchange rate provides less information about underlying fundamentals. Lastly, by increasing the importance of the noise component in exchange rate dynamics, the presence of strategic investors contributes to the heightened volatility of exchange rates compared to underlying fundamental factors, as fundamental factors exhibit lower volatility compared to noise shocks.

We empirically validate the theoretical predictions of the model using a panel of 10 currencies spanning from 2006 to 2016. We leverage the variation in market concentration across currencies and time to test the implications of our theory. For this purpose, we combine daily exchange rate data with currency-level concentration data from the U.S. Commodity Futures Trading Commission (CFTC). Our empirical results show that, consistent with the theoretical predictions, a currency traded in a market with a 10% higher share of strategic investors exhibits an 18% lower predictive power compared to the average predictive power in the data. Similarly, a currency traded in a market with a 10% higher share of strategic

investors exhibits an excess volatility ratio that is 12% higher than the average ratio.

We assess the potential impact of strategic investors on exchange rates by calibrating a benchmark model with strategic investors and comparing it to a competitive model. Under reasonable parameters, we show that the presence of strategic investors may significantly influence exchange rate disconnect. However, while strategic investors and heterogeneity in price impact may theoretically exacerbate the forward premium and excess volatility puzzles, they explain only a moderate portion of the puzzles. Abstracting from the heterogeneity in price impact increases the connection between fundamentals and exchange rates by 12%, while it accounts for a moderate increase in excess return predictability and exchange rate volatility of approximately 6%. In the sensitivity analysis, we show that the impact of investors' heterogeneity in price changes on exchange rate dynamics can substantially increase for stronger magnitudes of strategic behavior, supporting the idea that investor-level features may be quantitatively relevant for aggregate dynamics.

1.1 Related literature

Our work contributes to the microstructure approach to exchange rates by focusing on the heterogeneity of investors' price impact. Recent evidence from this literature highlight the importance of investor heterogeneity in understanding exchange rate dynamics and determination. For instance, the exchange rate determination puzzle, the excess predictability puzzle and the excess volatility puzzle can be explained by the rational confusion resulting from information heterogeneity among investors (Bacchetta and Van Wincoop, 2006; Candian and De Leo, 2022; Stavrageva and Tang, 2020). Furthermore, exchange rate behavior is linked to order flow, which, in turn, is associated with the heterogeneity among investors (Lyons et al., 2001; Evans and Lyons, 2006). However, despite extensive evidence that foreign exchange rate markets are highly concentrated and atomistic price-taking investors are hardly realistic, the literature has ignored the potential heterogeneity in price impact (Osler, 2014; Osler et al., 2016; Cochrane, 2015). A notable exception is the work in Corsetti et al. (2004) and Corsetti et al. (2002), which theoretically studies the role that large investors have in speculative attacks in the foreign exchange markets. Differently to them, we focus on exchange rate determination and puzzles by incorporating heterogeneity in price impact, drawing on the modeling approach of Kyle (1989) and Kacperczyk et al. (2018), which has not been previously applied in the context of exchange rate markets.

This paper contributes to the rich literature on the determination and dynamics of ex-

change rates in the presence of frictions. Prior work explores various types of frictions, including informational frictions (Evans and Lyons, 2002; Bacchetta and Van Wincoop, 2006), infrequent portfolio adjustment (Bacchetta and Van Wincoop, 2010, 2019), imperfect and frictional markets (Gabaix and Maggiori, 2015; He and Krishnamurthy, 2013). To the best of our knowledge, our work is the first to specifically focus on this aspect of the market structure – the presence of strategic investors and heterogeneity in price impact – for the determination of the exchange rate.

This paper also relates to the vast literature attempting to explain major puzzles in international economics, both theoretically and empirically. We contribute by providing a new rationale, based on strategic behavior and price impact, for the failure of macroeconomic fundamentals to predict exchange rates and the large volatility of the exchange rate relative to fundamentals (Meese and Rogoff, 1983; Obstfeld and Rogoff, 2000; Engel and Zhu, 2019). We also show that the presence of strategic behavior and excess predictability interact (Fama, 1984). Although we do not propose novel explanations for UIP deviations, the presence of strategic investors can account for currency level differences in UIP deviations. Moreover, we empirically study cross-currency differences in exchange rate puzzles and dynamics, which have been relatively unexplored, and find that different levels of price impact can explain cross-currency differences in a panel of 10 currencies.

The rest of the paper is organized as follows. Section 2 introduces the theoretical framework and explains the fundamental mechanism of strategic behavior. In Section 3, we discuss the main implications for the dynamics of the exchange rate and provide empirical evidence that supports the theoretical predictions. Finally, Section 4 presents the conclusion. Any proofs, derivations, and robustness analyses that were omitted can be found in the Appendices.

2 A Monetary Model with Strategic Investors

We propose a framework that incorporates strategic behavior in the spirit of Kyle (1989) and Kacperczyk et al. (2018) into a standard two-country, discrete time, general equilibrium monetary model of exchange rate determination (Mussa, 1982; Jeanne and Rose, 2002). We study the implications of strategic behaviour in the context of a monetary model, as it is well-suited for analyzing short-term exchange rate variations. In the baseline model, we assume that agents have rational expectations about the dynamics of the exchange rate to isolate

the key mechanism. In Section 3, we discuss a model extension that allow for dispersed information, following [Bacchetta and Van Wincoop \(2006\)](#).

2.1 Basic Set-up

There are two economies, Home and Foreign, both producing the same good. Variables referring to Foreign are indicated with a star. We assume that purchasing power parity holds, so that:

$$p_t = p_t^* + s_t,$$

where s_t is the log nominal exchange rate, p_t (p_t^*) the log price level in the Home (Foreign) country. The exchange rate is defined as the value of the foreign currency in term of domestic currency, and an increase in the exchange rate reflects an appreciation of the foreign currency. There are three assets: one-period nominal bonds issued by both Home and Foreign with interest rates i_t and i_t^* , respectively, and a risk-free technology with fixed real return r . The latter is infinitely supplied while bonds are in fixed supply in their respective currency. We follow [Bacchetta and Van Wincoop \(2010\)](#) and assume asymmetric monetary rules between the two countries. The Home central bank commits to a constant price level, $p_t = 0$, which implies that the domestic interest rate is equal to the risk free technology, $i_t = r$. On the other hand, the monetary policy in Foreign is stochastic, $i_t^* = -u_t$ where

$$u_t = \rho_u u_{t-1} + \sigma_u \varepsilon_t^u \quad \varepsilon_t^u \sim N(0, 1) \quad (1)$$

is the Foreign monetary policy shock. Thus, the interest rate differential is defined as

$$i_t - i_t^* = u_t + r,$$

implying that the dynamics of the exchange rate are solely influenced by the monetary policy of the Foreign country.⁵ In our model, we refer to a shock in the Foreign monetary policy as a fundamental shock.

There is a continuum of investors of mass one. We assume there are overlapping generations of investors that live for two periods and make only one investment decision. We abstract away from saving decisions by assuming that investors derive utility only from their

⁵[Bacchetta and Van Wincoop \(2010\)](#) specify a simplified Wicksellian rule of the form $i_t^* = \psi(p_t^* - \bar{p}^*) - u_t$ where ψ is set equal to zero, consistent with the low estimates of ψ reported by [Engel and West \(2005\)](#). [Bacchetta and Van Wincoop \(2010\)](#) show that an exogenous interest rate rule, as in our case, does not compromise the existence of a unique stochastic steady state for the exchange rate.

end-of-life wealth (Bacchetta and Van Wincoop, 2006, 2010). Investors in both countries are born with an exogenous endowment, ω , and have the possibility to invest in nominal bonds and the risk free technology. We assume that Foreign country is infinitesimally small, implying that the market equilibrium is determined by the investors located in the Home country. There are two type of investors: strategic (S) and competitive (C). A mass $1 - \lambda$ of investors consists of standard atomistic price-takers investors. The remaining segment, with size λ , consists of a finite number N of strategic investors. Each strategic investors has a positive mass, λ_i , with $\sum_i^N \lambda_i = \lambda$. Notably, strategic investors internalize their effect on asset prices, operating as an oligopoly.

Investor j maximizes mean-variance preferences over next period wealth, w_{t+1}^j , by allocating their initial endowment between domestic and foreign bonds:

$$\max_{b_t^j} \mathbb{E}_t^j(w_{t+1}^j | \Omega_t^j) - \frac{\rho}{2} \text{Var}_t^j(w_{t+1}^j | \Omega_t^j) \quad (2)$$

$$\text{s.t. } w_{t+1}^j = (\omega - b_t^j)i_t + (i_t^* + s_{t+1} - s_t)b_t^j, \quad (3)$$

where b_t^j represents the foreign bond holdings, ρ the rate of risk aversion and Ω_t^j the information set of investor j at time t . i_t and $i_t^* + s_{t+1} - s_t$ are the log-linearized returns of domestic and foreign bonds, respectively. Under PPP and the monetary policy assumptions above, we have that $p_t^* = -s_t$, implying that both returns are expressed in real terms. The only difference between the two assets is that the return on foreign bonds is stochastic.⁶ We assume that agents have symmetric rational expectations about the dynamics of the exchange rate, $\Omega_t^j = \Omega_t$.

Investors' demand schedule and portfolio allocation vary depending on their type. Strategic investors internalize the effects that their demand has on equilibrium prices (more precisely, on the equilibrium exchange rate), while competitive investors do not. In Appendix B, we show that the optimal demand for foreign bonds by investor j is as follows:

$$b_t^j = \begin{cases} \frac{\mathbb{E}_t s_{t+1} - s_t + i_t^* - i_t}{\rho \sigma_t^2}, & \text{for } j = C \\ \frac{\mathbb{E}_t s_{t+1} - s_t + i_t^* - i_t}{\rho \sigma_t^2 + \frac{\partial s_t}{\partial b_t^j}}, & \text{for } j = S \end{cases} \quad (4)$$

⁶ $p_t = 0$ implies $i_t = r$. Similarly, $p_t^* = -s_t$ implies that the return on foreign bonds, $i_t^* + s_{t+1} - s_t$, is expressed in real terms as well.

where σ_t^2 is the variance of the exchange rate change, $\text{Var}_t(s_{t+1} - s_t)$. We focus on a stochastic steady state where the variance σ_t^2 is time-invariant.

Investors' demand for foreign bonds depends positively on the expected excess return, $q_{t+1} \equiv \mathbb{E}_t s_{t+1} - s_t + i_t^* - i_t$. On the other hand, it depends negatively on the variance of the exchange rate, σ_t^2 , and on investors' risk aversion, ρ . Note that strategic behavior, captured by investors' own price impact $\frac{\partial s_t}{\partial b_t^{S,i}}$, reduces investors' demand of foreign bonds for every level of excess return. Given a total supply of foreign bond B , the price impact of a strategic investor i is

$$\frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i \rho \sigma_t^2}{B \rho \sigma_t^2 + (1 - \lambda)} > 0, \quad (5)$$

which is positive, increasing in the mass of the investor, λ_i , and decreasing in the fraction of atomistic investors $1 - \lambda$. The individual price impact becomes $\frac{1}{N} \frac{\lambda \rho \sigma_t^2}{B \rho \sigma_t^2 + (1 - \lambda)}$ in the case strategic investors are symmetric and have the same mass, $\lambda_i = \frac{\sum_i \lambda_i}{N} = \frac{\lambda}{N}$.⁷ The structure of the market determines the magnitude of the price impact and, consequently, the relevance of strategic behavior: the magnitude of the individual price impact is negatively affected by the number of strategic traders, N , and positively related to the size of the strategic segment, λ . Therefore, the price impact is larger in more concentrated markets characterized by a lower N and/or higher λ .⁸

In addition to strategic and competitive investors, we introduce another group of investors referred to as noise traders. As is standard, their presence allows to match key empirical moments of exchange rates, such as exchange rate volatility, disconnect and deviations from UIP (Kyle, 1989; Bacchetta and Van Wincoop, 2006, 2010). Following Bacchetta and Van Wincoop (2010), we assume that the demand of noise traders for foreign bonds is exogenous and given by:

$$X_t = (\bar{x} + x_t) \bar{W},$$

where $\bar{W}$ is the steady state aggregate financial wealth in the Home economy, $\bar{x}$ is a constant

⁷In our analysis, we focus on the case of symmetric strategic investors due to the unavailability of comprehensive investor-level market share data. Importantly, all qualitative predictions are not altered by the symmetry assumption. See Appendix B for the derivation of the analytic expression of the price impact.

⁸In our international portfolio model, strategic investors have a lower price impact on the equilibrium price of an asset compared to a closed-economy version. This is due to the presence of valuation effects on the supply of assets once denominated in domestic currency. By internalizing the effect that their demand has on the exchange rate, strategic investors also take into account how the value of the supply of foreign assets denominated in domestic currency varies when the exchange rate changes. This is reflected by the presence of B , the total supply of foreign assets, at the denominator of Equation (5). See Appendix B for additional details.

and x_t follows the following exogenous process:

$$x_t = \rho_x x_{t-1} + \sigma_x \varepsilon_t^x \quad \varepsilon_t^x \sim N(0, 1).$$

In the stochastic steady state, the demand for foreign assets absorbed by noise traders is equal to $\bar{x}\bar{W}$. Deviations from this steady state are driven by x_t , which is interpreted as a noise shock and is orthogonal to the fundamental shock u_t in Equation (1). Positive shocks to x_t increase the desire for foreign assets, leading the foreign currency to appreciate without movements in the interest rate differential.

Equilibrium and Basic Mechanism We derive an expression for the equilibrium exchange rate by combining the demand schedules of investors and the market clearing condition of the foreign bond market. The market clearing condition is given by:⁹

$$(1 - \lambda)b_t^C + \sum_i^N \lambda_i b_t^{S,i} + X_t = B e^{s_t}, \quad (6)$$

where the left hand side represents the total demand of foreign bonds from competitive investors, strategic investors and noise traders, and the right hand side represents the (constant) supply of foreign bonds, B , denominated in domestic currency.

We define the concept of equilibrium in our model as follow: for a history of fundamental and noise shocks $\{\varepsilon_t^u, \varepsilon_t^x\}_{t=0}^{-\infty}$, an equilibrium path is a sequence of portfolio allocations, $\{b_t^C, \{b_t^{S,i}\}_{i=1}^N\}$, and foreign bond price (exchange rate), $\{s_t\}$, such that investors optimally choose their portfolio allocation and the market clearing condition holds.

The model allows us to derive an explicit solution for the exchange rate s_t from the market clearing condition in Equation (6):

$$s_t = \underbrace{(1 - \mu) \left(\frac{\bar{x}}{b} - 1 \right)}_{\text{constant}} + \underbrace{\mu (\mathbb{E}_t s_{t+1} + i_t^* - i_t)}_{\text{fundamental}} + \underbrace{(1 - \mu) \frac{1}{b} x_t}_{\text{noise}}, \quad (7)$$

⁹The market clearing for the domestic bond is not explicitly considered because domestic bonds are perfectly substitutable with the risk free technology, which is infinitely supplied. Furthermore, in a monetary model, a market clearing condition for the money market would also be required. [Bacchetta and Van Wincoop \(2006\)](#) and [Bacchetta and Van Wincoop \(2010\)](#) assume that investors generate a money demand (independent of their portfolio decision) and that money supply accommodates it under the exogenous rule for interest rates. We do not explicitly model the money market in order to limit notation, leaving it in the background.

where $b = \frac{B}{W}$ and $\mu = \frac{1}{1+\Phi(\lambda, N)}$ with $\Phi(\lambda, N) = \frac{B\rho \text{Var}_t(s_{t+1})(1+B\rho \text{Var}_t(s_{t+1})-\lambda\frac{N-1}{N})}{(1+B\rho \text{Var}_t(s_{t+1})-\lambda\frac{N-1}{N})-\frac{\lambda^2}{N}}$. The exchange rate follows a forward looking auto-regressive process with drift, where the constant term depends on a set of parameters and the stochastic component depends on future fundamental and noise shocks. By further manipulating Equation (7), it can be shown that the exchange rate s_t can be written as follows:

$$s_t = \mu \sum_{k=0}^{\infty} \mu^k \mathbb{E}_t(i_{t+k}^* - i_{t+k}) + \frac{1-\mu}{b} \sum_{k=0}^{\infty} \mu^k \mathbb{E}_t(x_{t+k}). \quad (8)$$

The exchange rate is a weighted average of current and expected future fundamental shocks $(i_{t+k}^* - i_{t+k})$ and noise shocks (x_{t+k}) . The weight μ quantifies the amount of information about the fundamental conveyed by the exchange rate. Notably, the informativeness of the exchange rate decreases when strategic investors operate in the foreign bond market (higher λ or lower N imply higher Φ and, thus, lower μ). When there is a higher proportion of strategic investors (higher λ) or a lower number of strategic traders (lower N), investors' demand declines because of the stronger price impact. Therefore, the demand from noise traders becomes relatively more important in determining the exchange rate.^{10 11}

2.2 Theoretical Predictions

We illustrate the theoretical implications of strategic behavior for exchange rate dynamics. Specifically, we show that the presence of strategic investors may increase exchange rate volatility, exacerbate the disconnect between the exchange rate and underlying fundamentals, and lead to larger deviations from uncovered interest parity (UIP). Appendix B provides the proofs for each proposition.

Response to Shocks In the model, an increase in the exchange rate arises due to either a positive noise shock or a negative fundamental shock. A positive noise shock, interpreted as either an increase in demand or a decrease in the supply of foreign assets, reduces the residual demand for these assets. This leads to a rise in the price of foreign assets and an

¹⁰When traders recognize that the residual supply curve is upward-sloped, quantities are restricted and less elastic. Therefore, prices become less informative. This aligns with the key intuition from Kyle (1989).

¹¹The informativeness parameter, μ , relates to the magnification factor in Bacchetta and Van Wincoop (2006). In their work, information dispersion among investors reduces the information content of exchange rates by amplifying the impact of noise traders. As in their work, the behavior of the parameter μ plays a crucial role in the amplification mechanism examined here.

appreciation of the exchange rate, despite no change in the underlying fundamentals. As the exchange rate increases, the excess return falls below its steady state, causing investors to reduce their foreign asset holdings and rebalance their portfolios in favor of domestic assets. Conversely, a negative fundamental shock, caused by contractionary monetary policy in the foreign country, reduces the interest rate differential, and directly increases the excess return on holding foreign assets. This in turn boosts investors' demand for foreign assets, leading to an appreciation of the foreign currency.

Proposition 1 (Response to Shocks). *An increase in the size of strategic investors amplifies the response of the exchange rate to noise shocks while dampening the response to fundamental shocks.*

Proposition 1 states that the response of the exchange rates are amplified in response to a noise shock and dampened in response to a fundamental shock when strategic investors are present in the market. To see this, consider the law motion of the exchange rate in Equation (7). Because the fundamental and noise processes are AR(1) with persistences ρ_u and ρ_x , $E_t(u_{t+k}) = \rho_u^k u_t$ and $E_t(x_{t+k}) = \rho_x^k x_t$. Substituting and summing the geometric series gives the closed-form solution

$$s_t = -\frac{\mu}{1 - \mu\rho_u}u_t + \frac{(1 - \mu)}{(1 - \mu\rho_x)b}x_t \quad (9)$$

The exchange rate's response to one standard deviation shock in noise ε_t^x and fundamental ε_t^u components is given by:

$$\mathbb{E}(\Delta s_t | \varepsilon_t^u = -1) = \frac{\mu\sigma_u}{1 - \mu\rho_u}, \quad \mathbb{E}(\Delta s_t | \varepsilon_t^x = 1) = \frac{(1 - \mu)\sigma_x}{(1 - \mu\rho_x)b}. \quad (10)$$

Since μ is decreasing (increasing) function of the size of strategic investor λ in the market, then the response of the exchange rate to a unit shock in fundamental is dampened while noise shock are amplified as λ increases. Figure 7 in Appendix C provides a graphical representation on the impulse response functions.

Economically, this occurs because strategic investors, who internalize the negative impact their trades have on prices, consistently respond less aggressively to aggregate shocks, which affects the sensitivity of foreign asset demand. Therefore, in a world where investors behave strategically, the decline in demand for foreign assets in response to a noise shock is less pronounced compared to a competitive benchmark. The smaller decline in the demand for foreign bonds, due to strategic behavior, exerts additional upward pressure on the price of foreign bonds, thereby amplifying the effect of noise shocks on the exchange rate.

Conversely, in response to a fundamental shock, strategic investors increase their demand for foreign assets less than they would in a competitive market, again making total demand less sensitive to fundamental shocks. In this strategic environment, investors internalize their price impact, leading to a smaller increase in their holdings of foreign assets. As a result, the price of foreign assets rises less than it would in a competitive market, dampening the effect of the fundamental shock on the exchange rate.

Excess Return Predictability An empirically robust evidence in exchange rate dynamics is the predictability of excess returns, commonly referred to as deviations from the Uncovered Interest Parity (UIP) (Fama, 1984). Our model predicts systematic deviations from UIP due to a non-zero net supply of foreign assets, regardless of the presence of strategic investors.

Proposition 2 (Excess Return Predictability). *Excess returns is more predictable as the size of strategic investors in the market increases.*

Proposition 2 states that strategic behavior amplifies these UIP deviations compared to a competitive market. To illustrate this, define the 1-period excess return $q_{t+1} = s_{t+1} - s_t - (i_t - i_t^*)$, and consider the corresponding one-period Fama regression:

$$q_{t+1} = \alpha + \beta^{\text{Fama}}(i_t - i_t^*) + \nu_t, \quad (11)$$

where the coefficient, β^{Fama} , reflects deviations from the UIP theory. If UIP holds, the coefficient of the Fama regression is expected to be zero. However, in empirical studies, the Fama coefficient is systematically different from zero and negative. Based on Equation (11), the coefficient β^{Fama} is:

$$\beta^{\text{Fama}} \equiv \frac{\text{Cov}(q_{t+1}, i_t - i_t^*)}{\text{Var}(i_t - i_t^*)} = -(1 - \mu) \frac{1}{1 - \mu \rho_u} \in (-1, 0), \quad (12)$$

which is negative, in line with empirical findings, and decreases as strategic behavior increases. In other words, when there are strategic investors, changes in the interest rate differential have a stronger impact on the excess return, and consequently on the risk premium.

In our theoretical framework, as strategic investors' positions become less elastic to aggregate shocks, the market requires larger movements in the risk premium to incentivize investors to take larger positions. An increase in the size of strategic investors, λ , reduces

market risk appetite, further necessitating a higher risk premium to absorb the net supply of foreign assets. This makes the excess return more responsive to changes in fundamentals, increasing its predictability. We can re-write the excess return as:

$$\mathbb{E}_t q_{t+1} \approx \Phi (Be^{s_t} - X_t) \quad (13)$$

The right-hand side represents the deviation from UIP, which can be interpreted as the risk premium required by investors for holding a foreign asset. The risk premium depends on two components: the net supply of foreign assets and the size of non-competitive investors, captured by Φ , which increases with the size of strategic investors, λ . The first component drives the deviation from UIP in the model, while the second component, Φ , amplifies the UIP deviation.¹²

Exchange Rate Excess Volatility There is extensive evidence showing that exchange rates exhibit higher volatility compared to fundamentals, which is commonly referred to as the “excess volatility puzzle” (Obstfeld and Rogoff, 2000; Engel and Zhu, 2019). This puzzle provides evidence of the significant role of non-fundamental components in explaining exchange rate dynamics in the data.

Proposition 3 (Exchange Rate Excess Volatility). *Given standard parametrization, the noise component accounts for a large share of exchange rate variance. Thus, an increase in the size of strategic investors makes the exchange rate more volatile relative to fundamentals.*

Proposition 3 states that if variations in exchange rates are mainly driven by noise, strategic investors contribute to the excess volatility of the exchange rate relative to fundamentals. By manipulating Equation (8), we derive an expression for the unconditional variance of the exchange rate as a combination of the variances of both fundamental and noise shocks:

$$\text{Var}(s_t) = \frac{\mu^2}{(1 - \mu\rho_u)^2} \left[\frac{1}{1 - \mu^2} + \frac{\rho_u^2}{1 - \rho_u^2} \right] \sigma_u^2 + \frac{(1 - \mu)^2}{(1 - \mu\rho_x)^2 b^2} \left[\frac{1}{1 - \mu^2} + \frac{\rho_x^2}{1 - \rho_x^2} \right] \sigma_x^2. \quad (14)$$

¹²Appendix B generalizes the result in Proposition 2 showing that the Fama coefficient is monotonically increasing in the time horizon k , and approaching zero for $k = \infty$. Therefore, our model does not explain the predictability reversal puzzle documented in Bacchetta and Van Wincoop (2010) and Engel (2016), which shows that expected excess returns reverse sign at longer horizons. This limitation is anticipated, as our focus is on an amplification mechanism without portfolio adjustment frictions (Bacchetta and Van Wincoop, 2010, 2019).

We compute the excess volatility of the exchange rate as the ratio of the volatility of the exchange rate in Equation (14) to the volatility of the fundamental, $\frac{\sigma_u}{\sqrt{1-\rho_u^2}}$ (Engel and Zhu, 2019). This is increasing in λ if and only if:

$$\frac{\text{Var}(x_t)}{\text{Var}(i_t - i_t^*)} \frac{1}{b^2} > \left[\frac{(1 + \mu^2 \rho_x)(1 - \rho_x)}{\mu(1 + \mu \rho_u)(1 - \mu^2) + \mu^3(1 - \rho_u^2)} \frac{(1 - \mu \rho_u)^2(1 - \mu)^2}{(1 - \mu \rho_x)^2} \right]^{-1}. \quad (15)$$

This suggests that the unconditional variance of the exchange rate increases with the share of strategic investors when the variance of the noise shock is sufficiently high relative to the variance of the fundamental process.

Intuitively, this occurs because the presence of strategic investors reduces the informativeness of the exchange rate, placing relatively more weight on the noise component. Given standard parametrization, the noise component is more volatile than the fundamental component.¹³ Thus, when exchange rates are primarily driven by noise, strategic investors contribute to the increased volatility observed in the exchange rate.

Exchange Rate Disconnect One of the most well-established empirical findings in exchange rate dynamics is the disconnect between exchange rates and fundamentals (Meese and Rogoff, 1983; Cheung et al., 2005; Rossi, 2013). Specifically, variations in the interest rate differential explain only a small portion of the changes in exchange rates.

Proposition 4 (Exchange Rate Disconnect). *Given standard parametrization, the disconnect between the exchange rate and fundamentals increases in the presence of strategic investors.*

To measure the disconnect, we assess the explanatory power of the following regression equation:

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \nu_{t+1}, \quad (16)$$

where $i_t - i_t^*$ represents the fundamental driver of the one-period exchange rate change $s_{t+1} - s_t$. The proportion of variance in exchange rate changes explained by fundamentals is:

$$R^2 = \frac{\text{Var}(i_t - i_t^*)}{\text{Var}(\Delta s_{t+1})} [1 + \beta^{Fama}]^2 \quad (17)$$

The R^2 coefficient depends on two factors: the ratio of the variance of the fundamental to

¹³The quantitative assessment in Appendix C show that, monotonicity is satisfied for any reasonable calibration, although the effect of strategic behavior is not necessarily monotonic from a theoretical perspective.

the variance of exchange rate changes, and the Fama coefficient from Proposition 2. Given standard parametrization, an increase in the size of strategic investors raises exchange rate volatility, reducing the first term in Equation (17). Simultaneously, an increase in the size of strategic investors makes the exchange rate more predictable, reducing the second term. The two effects together unambiguously make the exchange rate more disconnected from fundamentals. The main intuition relies again on the fact that strategic investors reduce the informativeness of the exchange rate, placing relatively more weight on the noise component.

2.3 Testing Predictions

We leverage the heterogeneity in market concentration across currencies to test the implications of our theory. The model delivers three distinct testable relationships between exchange rate dynamics and the level of strategic behavior: (i) currencies with a larger presence of strategic investors are more predictable; (ii) higher levels of strategic behavior result in greater excess volatility in exchange rates; and (iii) the disconnect between exchange rates and fundamentals increases with the level of strategic behavior. We test model’s predictions using a set of 10 currencies merged with the U.S. CFTC transaction data, available since June 2006 to December 2016.¹⁴

We use the concentration ratio of the top eight investors, as reported by the U.S. CFTC, as our proxy of strategic behavior in the foreign exchange market (λ).¹⁵ We correlate this information with time-varying metrics of exchange rate disconnect, predictability and excess volatility. We utilize a 2-year rolling window with monthly exchange rate data to create time-varying indexes for exchange rate predictability, disconnect and excess volatility. We measure exchange rate predictability as the coefficient on the interest rate differential from the Fama regression in Equation (11), exchange rate disconnect using the R^2 from the regression in Equation (16), and excess volatility as the ratio of exchange rate volatility (from Equation (14)) to the volatility of the interest rate differential. The panel nature of our dataset enable us to incorporate currency and year fixed effects, mitigating potential concerns regarding spurious correlation and strengthening the validity of the empirical evidence.¹⁶

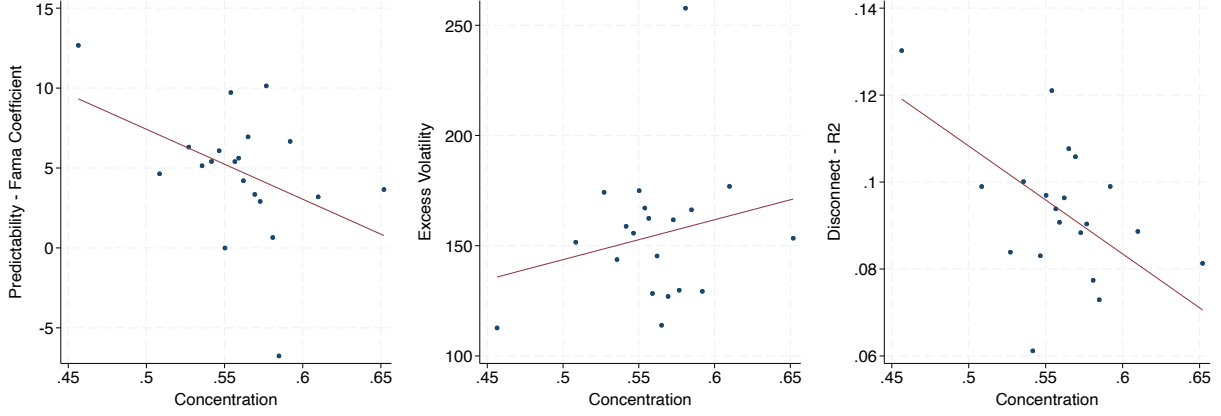
Figure 2 provides evidence that are consistent with the predictions of our theoretical

¹⁴We exclude the South African Rand from our analysis due to a limited number of time observations in the CFTC dataset. To reduce noise in weekly transactions, we aggregate the data to a monthly level.

¹⁵The assumption is that variations in concentration ratios between currencies may reflect differences in the presence and influence of traders who engage in strategic trading.

¹⁶The results remain unchanged when we increase the rolling window size to 3 and 4 years, and using different proxies for strategic behavior, such as the number of active traders.

Figure 2: Testing Model Predictions



Notes: The figure plots the positive relationship between the level of strategic behavior and the excess return predictability (left panel), the excess volatility (central panel), and the disconnect (right panel) of the exchange rate in the actual data. Concentration is the share of open interest held by the top eight traders in the future FX market. Data is from the U.S. CFTC spanning from 2006 to 2016. Exchange rate predictability is represented by the regression coefficient from the Fama regression in Equation (11). The exchange rate disconnect is measured using the R^2 from the regression in Equation (16), while excess volatility is calculated as the ratio of exchange rate volatility (from Equation (14)) to the volatility of the interest rate differential. To measure predictability, excess volatility, and disconnect, we use 2-year rolling window regressions based on monthly average exchange rate data. The resulting data are demeaned at the currency and year levels, and the Fama coefficient, excess volatility ratio, and disconnect values are winsorized at the 1% level. The South African Rand is excluded from the set of 11 currencies. The estimated coefficients are reported in Table 5 in Appendix D, while Appendix A provides further details on the data used.

framework. The left panel shows that, on average, currencies with more concentrated trading flows, held by fewer investors, tend to be more predictable. The middle panel documents a strong, positive, and statistically significant relationship between our measure of strategic behavior in exchange rate markets and the excess volatility of exchange rates. Finally, the right panel reveals that as the presence of strategic investors in the market increases, currencies become more disconnected from fundamentals, as indicated by the decreasing estimated R^2 .

Table 5 in Appendix B reports the estimated coefficients along with the corresponding standard errors, clustered at the country level. We find that a currency traded in a market with a 10% higher concentration ratio exhibits an excess volatility ratio that is approximately 12% higher than the sample average. Similarly, a currency traded in a market with a 10% higher concentration ratio exhibits 18% lower predictive power compared to the average R^2 observed in the sample.

3 Quantitative Assessment

To what extent can heterogeneity in price impact account for exchange rate puzzles? We approach this question through the lens of the framework developed in Section 2. While its analytical tractability affords sharp comparative statics linking the size of the strategic segment to the manifestation of these puzzles, we are mindful that such parsimony is not without cost. The model’s deliberately stylized structure necessarily circumscribes its scope, and thus the ensuing exercise is best interpreted as a disciplined theoretical benchmark rather than a definitive quantification of the mechanism’s empirical relevance.

Moreover, a precise quantitative answer would require transaction-level data and information on the identity and positioning of individual players in the foreign exchange market, which is not available to us. In particular, two parameters are difficult to discipline directly from the data: the size of the strategic segment λ and the coefficient of relative risk aversion ρ . Concentration ratios provide a natural empirical proxy for λ but are an imperfect measure of strategic behavior — large investors may refrain from strategic action because of regulatory scrutiny, while smaller but sophisticated traders may behave strategically despite a smaller market share. Risk aversion, in turn, is unobservable and the literature offers a wide range of values. Rather than committing to a single point estimate, we adopt a benchmark calibration disciplined by CFTC concentration data and existing studies, and then vary λ and ρ across plausible ranges to map out the contribution of the mechanism under alternative assumptions. This approach reflects the inherent uncertainty about the true level of strategic behavior in foreign exchange markets and provides a range of estimates rather than a single number.

3.1 Range of Quantitative Estimates

We calibrate our framework to a panel of 18 currencies, run counterfactuals against a competitive benchmark, and quantify the contribution of the micro-mechanism to the exchange rate disconnect, the forward premium puzzle, and exchange rate volatility through the lens of a monetary macro model.

Calibration. The benchmark parametrization is reported in Table 6. We calibrate the model using monthly exchange rate data on 18 currencies against the U.S. Dollar from 1993

to 2019.¹⁷ Without loss of generality we set $\bar{r} = 0$, which implies that the interest rate differential reduces to $i_t - i_t^* = u_t$. We assume that covered interest rate parity holds and compute the one-month interest rate differential as the difference between the log one-month forward and the log spot exchange rate. We model the fundamental as an AR(1) process and estimate its volatility and persistence currency by currency, calibrating $\sigma_u = 0.005$ and $\rho_u = 0.85$ to match the cross-currency averages. The variance of the one-period exchange rate change, σ_t , is calibrated to its empirical counterpart, 0.028.

The noise process x_t is calibrated to match exchange rate dynamics. We set $\rho_x = 0.9$, high enough to keep the exchange rate sufficiently close to a random walk, and choose σ_x to match the empirical volatility of the one-period exchange rate change conditional on the calibrated market-structure parameters. As Equation (8) makes clear, exchange rate dynamics depend jointly on the noise process and on the market structure, so the noise volatility required to match a given empirical target depends on λ and N . We therefore first calibrate λ and N , and then estimate σ_x residually.

We use data from the U.S. Commodity Futures Trading Commission to calibrate the market-structure parameters. We set $N = 8$ and $\lambda = 0.572$, the latter being the average concentration ratio of the top eight reportable traders in the FX futures market over 2006–2016. Given these values, we obtain $\sigma_x = 0.133$. We follow [Bacchetta and Van Wincoop \(2019\)](#) and set $b = 0.33$ — implying that foreign assets account for roughly one third of domestic financial wealth — and the relative risk aversion coefficient at $\rho = 50$.

Results. We assess the impact of strategic behavior by comparing simulated moments under the benchmark parametrization to those generated by an otherwise identical competitive economy in which $\lambda = 0$ (all investors are atomistic price takers). For each model specification we run 1,000 simulations of 6,000 periods with a 3,000-period burn-in, and report the moments summarizing the exchange rate response to fundamental and noise shocks, the Fama coefficient, exchange rate volatility, and the disconnect R^2 . Table 1 reports the results under different parametrization in percentage deviation from a competitive model. Point estimates of the results are in the Appendix C.

Under the benchmark, strategic behavior amplifies the response to noise shocks more than it dampens the response to fundamental shocks. A one-standard-deviation noise shock raises the exchange rate by 14.66% in the competitive economy and by 15.51% in the strategic benchmark, a 5.8% increase. The corresponding dampening of the fundamental response is

¹⁷The currencies are listed in Appendix A.

Table 1: Summary of the Quantitative Results

Moments from Simulated Data (% Deviation from Competitive Model)	(1) Response to Shock Noise (Fundamental)	(2) Fama Coefficient	(3) Exchange Rate Volatility (St.D.)	(4) Exchange Rate Disconnect
Benchmark Case	0.058 (0.017)	-0.064	0.055	-0.122
Size Strategic Investors				
Low ($\lambda/N = 5\%$)	0.021 (0.006)	-0.024	0.02	-0.047
High ($\lambda/N = 9\%$)	0.128 (0.038)	-0.144	0.123	-0.247
Risk Adversion				
Low ($\rho = 30$)	0.068 (0.011)	-0.073	0.061	-0.117
High ($\rho = 70$)	0.05 (0.021)	-0.057	0.049	-0.118
Periods per Simulation	6000	6000	6000	6000
Burn-in Periods	3000	3000	3000	3000
Number of Simulations	1000	1000	1000	1000

Notes: The Table reports the impact of strategic investors on exchange rate dynamics as percentage deviations from the competitive model using simulated data (e.g., 0.01 unit indicates a 1% deviation from the competitive model). For comparison, the first row presents the impact of strategic investors on exchange rate moments in the benchmark case. Each column reports percentage deviations from the competitive model. Column (1) shows the amplification or dampening of the exchange rate response to a one-standard-deviation noise and fundamental shock. Column (2) reports the percentage deviation in the Fama regression beta. Column (3) reports the percentage deviation in exchange rate volatility, measured by the standard deviation, and Column (4) provides the percentage deviation in the estimated R^2 from the disconnect regression. The results are obtained from 1,000 simulations, with each iteration running for 6,000 periods and a burn-in of 3,000 periods. All the other parameters are held constant across scenarios; see Table 6 for details.

an order of magnitude smaller (approximately 2%). The asymmetry reflects two features of the calibration: the noise process is more persistent than the fundamental ($\rho_x = 0.9$ vs. $\rho_u = 0.85$), and the noise component enters with a coefficient scaled by $1/b$, which amplifies its weight given the calibrated home bias. The presence of strategic investors moves the Fama coefficient from -0.204 to -0.217 (roughly 6% more negative), increases exchange rate volatility by approximately 6%, and reduces the disconnect R^2 from 0.083 to 0.073, a 12% decrease. Under this calibration, the mechanism contributes meaningfully to the exchange rate disconnect and modestly to the forward premium and excess volatility puzzles.

The benchmark sets λ at the cross-currency average. The contribution of the mechanism scales nontrivially with this parameter. Reducing the average market share of a strategic investor from $\lambda/N \approx 7\%$ to 5% shrinks the noise amplification from 5.8% to 2.1% and the disconnect contribution from 12% to 4.7%. Conversely, raising the average market share to 9% — still well within empirical concentration ranges for non-major USD pairs (Figure 1) — amplifies the noise response by 12.8%, makes the Fama coefficient 14.4% more negative,

raises volatility by 12.3%, and accounts for a 24% reduction in the disconnect R^2 . Figure 9 in Appendix C plots the moments across the full range of λ . The response to noise shocks rises from approximately 0% in the competitive limit to nearly 38% as $\lambda \rightarrow 1$, and the disconnect contribution increases monotonically over the same range. The mechanism’s contribution to exchange rate puzzles therefore depends critically on the assumed size of the strategic segment, and is substantially larger at the concentration levels observed for non-major currencies than at the all-currency average used in the benchmark.

Finally, lowering risk aversion to $\rho = 30$ raises the noise response by 6.8% relative to the competitive model and increases volatility by 6.1%; raising risk aversion to $\rho = 70$ moderates both effects. The intuition is that risk-tolerant strategic investors are willing to take larger positions, which makes their internalized price impact more consequential for equilibrium prices.¹⁸

3.2 Taking Stock

The quantitative exercise yields two main findings. First, at the average level of strategic behavior observed in our CFTC sample, the mechanism accounts for a modest portion of the exchange rate disconnect and only a limited share of the forward premium and excess volatility puzzles. While these effects are non-negligible, we do not view them as constituting a resolution of any of the puzzles. Second, the contribution scales nonlinearly with the size of the strategic segment. At concentration levels observed for non-major USD currency pairs, the mechanism becomes quantitatively significant—particularly for the disconnect—highlighting the importance of accurately measuring market concentration.

We emphasize, however, that this exercise is subject to important limitations. The CFTC data pertain to the U.S. FX futures market and may not provide a comprehensive measure of concentration in the broader FX market. Moreover, while the model delivers sharp predictions, it abstracts from several features that may be quantitatively relevant, such as intermediary frictions that could interact with strategic behavior in important ways (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021).¹⁹

¹⁸This pattern operates through the equilibrium informativeness parameter μ rather than through the standard risk-premium channel emphasized in Itskhoki and Mukhin (2021), where higher risk aversion makes excess returns more sensitive to flows. In our setup, higher ρ raises $\Phi(\lambda, N)$, but it also raises $B\rho\text{Var}_t(s_{t+1})$ in both the strategic and competitive benchmarks, so the differential impact of strategic behavior on dynamics is muted. The two channels are not in conflict; they act on different objects.

¹⁹In Appendix D, we show that within our framework strategic behavior generates effects on exchange rate dynamics that are quantitatively comparable to those arising from a well-established source of het-

4 Conclusion

The heterogeneity in price impact and concentration in the foreign exchange rate markets may play an important role in understanding exchange rate dynamics.

In this paper, we explore the implication of strategic behavior within a simple monetary model of exchange rate determination. We show that strategic behavior reduces the informativeness of the exchange rate by amplifying the response to non-fundamental shocks while dampening the response to fundamental shocks. As a result, heterogeneity in price impact helps to explain the weak empirical link between fundamentals and exchange rates, as well as the excess volatility observed in exchange rate movements.

Although our model is stylized to derive fundamental insights and analytic results, we provide empirical evidence supporting the theoretical predictions using a panel of 10 currencies. Furthermore, we extend the theoretical framework by including a competing dimension of investors' heterogeneity, namely information dispersion.

This paper represents a step forward in incorporating microstructure institutions in the analysis of exchange rate dynamics. Our framework is tractable and can be integrated into macro models of exchange rate determination. As shown in previous literature, the introduction of investor heterogeneity qualitatively and quantitatively alters conclusions regarding optimal monetary and exchange rate policies. It also calls for additional efforts in documenting and studying investors' heterogeneity in foreign exchange rate markets.

erogeneity—information dispersion. While both mechanisms share similar directional implications, they operate through distinct channels. The comparable magnitudes further underscore the relevance of market concentration in foreign exchange markets.

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Appendix

A Empirics

A.1 Data

We use three main sources of data:

- We use data on 18 currencies from December 1993 to December 2019. The currencies considered are: Euro, Japanese Yen, Argentinian Peso, Brazilian Real, Canadian Dollar, Swiss Franc, Australian Dollar, Chilean Peso, Indian Rupee, Mexican Peso, British Pound, South African Rand, Russian Ruble, Swedish Krona, Turkish Lira, New Zealand Dollar, Singapore Dollar, Norwegian Krone. The panel is not balanced.

We obtain data for the spot and one-month forward exchange rates at a daily frequency from Datastream and Thompson Reuters. All exchange rates are defined against the US Dollar. To calculate the one-month interest rate, we took the difference between the logarithm of the one-month forward exchange rate and the logarithm of the spot exchange rate. We then computed monthly averages for the spot exchange rates and the one-month interest rate differentials.

- We use data from the U.S. Commodity Futures Trading Commission (CFTC) on investors' currency positions. The U.S. Commodity Futures Trading Commission (CFTC) data provides detailed information on several aspects within the currency futures market, including net open interest positions held by asset managers, institutional investors, and leveraged funds, as well as measures of concentration and the number of reportable traders. Data is reported on a weekly basis and spans the years 2006 to 2016 for 11 currency pairs. These pairs include both major and non-major USD currency pairs, reflecting the diversity of assets traded within the currency futures market. Major currency pairs typically involve the U.S. dollar and another major global currency, such as the Euro or Japanese Yen, while non-major pairs may involve currencies from emerging markets or smaller economies. Table 2 in Appendix A reports summary statistics on key variables. Figures ??, 3 and 4 in Appendix A respectively show the net open positions, the concentration ratio, and the number of reportable traders in the future FX market per currency and trader group.

- We use data on exchange rate expectations from the ECB Professional Forecasters survey. The survey runs at quarterly frequency since 2002Q1 until 2020Q4. It provides information on the expectations of professional forecasters regarding the euro-dollar exchange rate at different time horizons, including the current quarter and one to four quarters ahead. The dataset includes exchange rate forecasts from approximately forty professional forecasters.

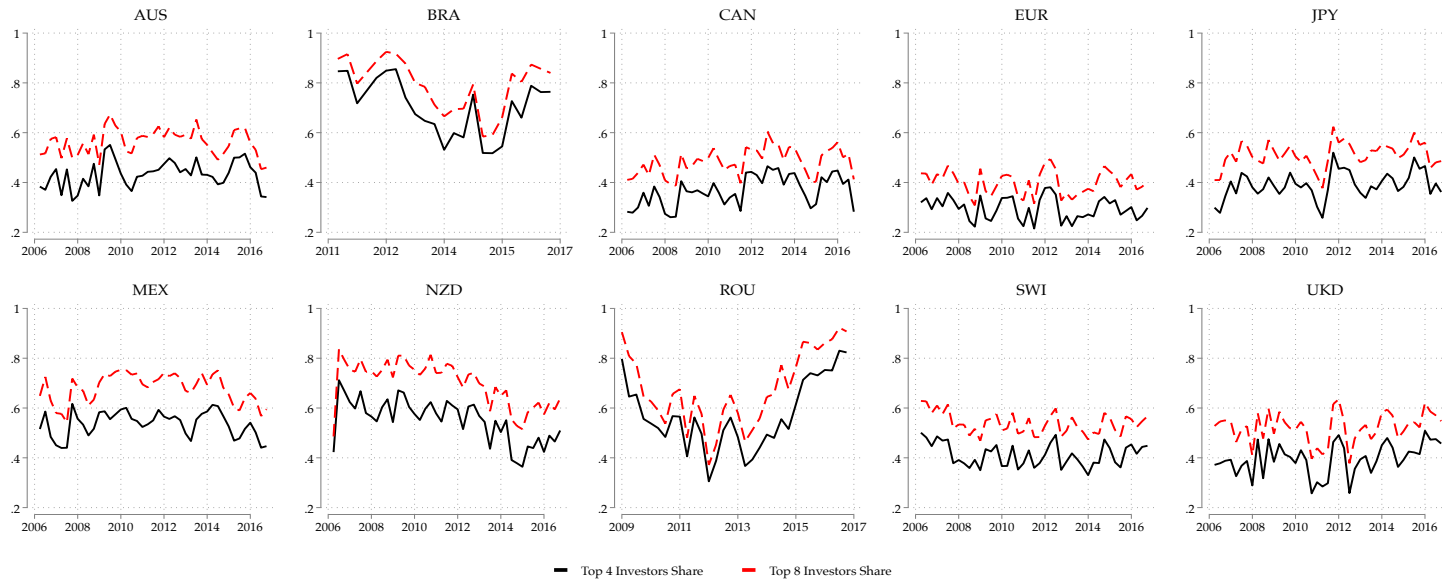
A.2 Additional Figures and Tables

Table 2: Summary Statistics - Key Variables from CFTC dataset

	Currency										Total
	AUS	BRA	CAN	EUR	JPY	MEX	NZD	ROU	SWI	UKD	
Imbalances (Mil USD): Intermediaries	-16.180	-6.440	-12.658	28.322	18.114	-32.066	-6.517	-7.947	5.379	22.880	-0.136
Institutional Investors	-7.799	2.060	2.166	-5.080	11.954	17.415	-1.776	1.024	-0.632	-22.889	-0.544
Hedge Funds	24.303	3.980	-1.949	-26.026	-20.759	15.297	7.938	5.004	-3.839	10.270	1.152
Others	-4.234	0.303	5.850	9.987	2.196	-2.221	-0.283	0.969	0.263	-7.536	0.530
Concentration: Top 4 (Net)	0.433	0.699	0.368	0.295	0.392	0.537	0.553	0.566	0.411	0.396	0.448
Top 8 (Net)	0.564	0.795	0.484	0.399	0.513	0.680	0.704	0.676	0.537	0.523	0.572
Number: Intermediaries	7.788	4.693	7.448	13.411	8.921	7.075	6.764	6.327	6.180	7.272	7.830
Institutional Investors	6.115	0.084	5.428	15.684	7.773	6.200	2.744	0.000	1.974	7.333	6.442
Hedge Funds	15.614	6.129	13.881	25.069	19.697	16.122	9.867	4.883	8.898	16.649	14.738
Others	5.361	2.340	7.266	12.580	6.352	6.866	4.201	0.482	0.199	5.114	5.952

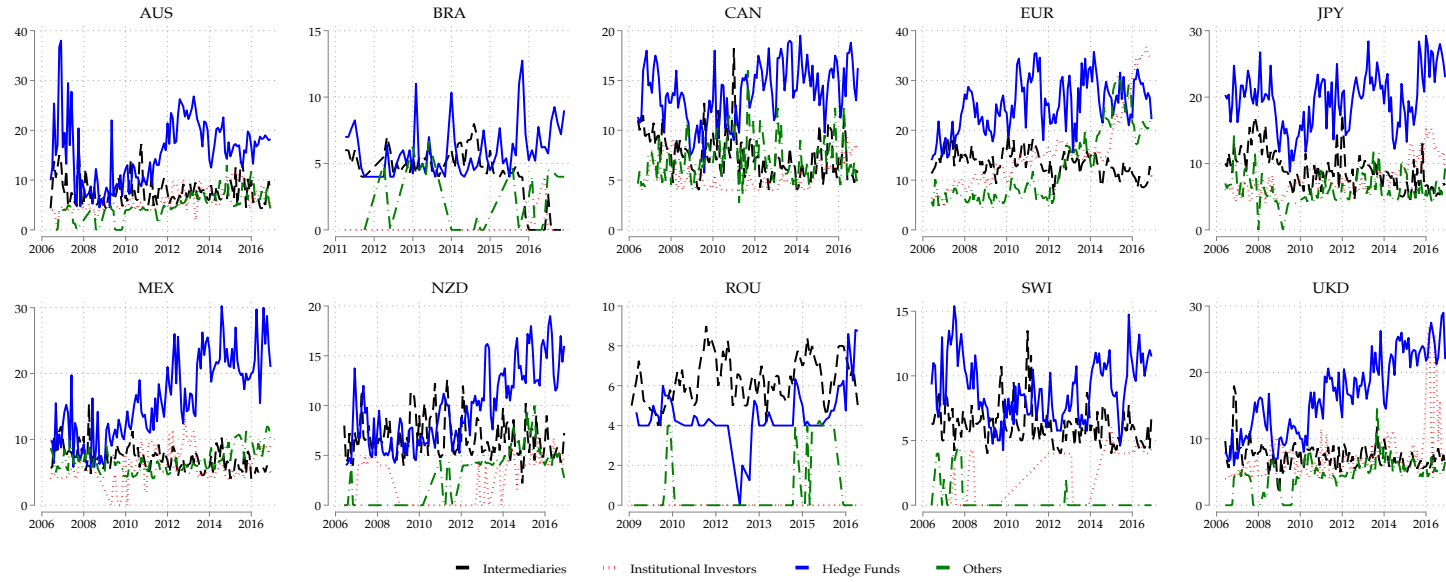
Notes: Report the means of the main variables in the CFTC dataset by currency pair. Net open interest positions are in millions of dollars (\$), concentration ratios are expressed in percentages, and the number of traders is in count. The reported mean statistic is calculated based on a panel of weekly observations spanning from 2006 to 2018.

Figure 3: Concentration Ratios by Currency Pairs



Notes: The figure shows the average percentages of open interest held, referred to as Concentration Ratios, by the largest four (black line) and eight (red line) reportable traders in the future FX market by currency pair. These concentration ratios are based on 'Net Position' and are calculated after offsetting each trader's equal long and short positions. The data is sourced from the U.S. Commodity Futures Trading Commission (CFTC) and spans from 2006 to 2016. Data are quarterly averages for each currency pair. Appendix A provides additional details regarding the data used.

Figure 4: Number of Traders by Currency Pairs



Notes: The figure shows the numbers of reportable traders in the future FX market by currency pair. For each currency pair, we report the number of Dealers (black dashed line), Institutional Investors (red dotted line), Hedge Funds (blue line) and Other Reportable Traders (green dashed line). The data is sourced from the U.S. Commodity Futures Trading Commission (CFTC) and spans from 2006 to 2016. Data are quarterly averages for each currency pair. Appendix A provides additional details regarding the data used.

Table 3: Summary Statistics - Exchange Rates

Panel A. Monthly frequency

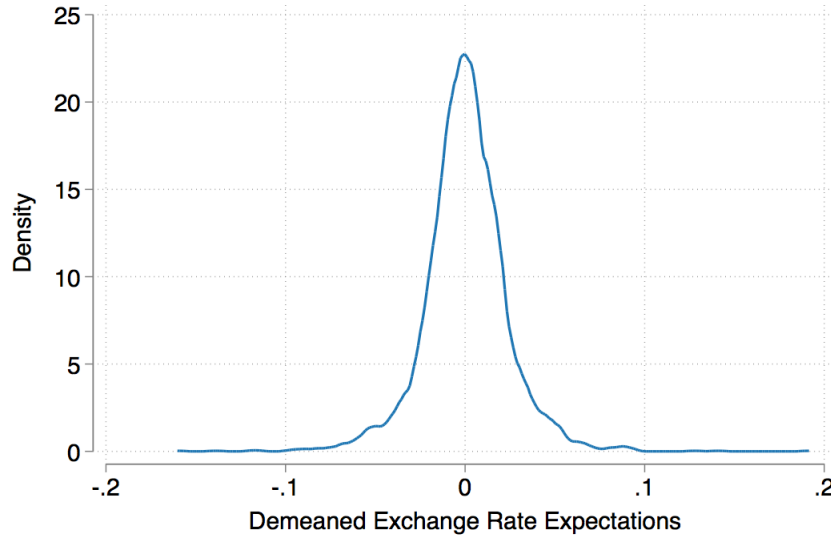
Variables	Mean	SD	Min	Max	N
Exchange rate return (in %)	0.334	0.548	-0.130	1.910	18
Exchange rate level volatility (StD)	0.344	0.353	0.115	1.367	18
Volatility of exchange rate changes (StD)	0.029	0.012	0.013	0.055	18
Fama coefficient (UIP)	-0.603	1.317	-2.688	2.357	18
R-Squared (Exchange rate disconnect)	0.030	0.065	0.000	0.265	18

Panel B. Weekly frequency

Variables	Mean	SD	Min	Max	N
Exchange rate return (in %)	0.077	0.127	-0.030	0.442	18
Exchange rate level volatility (StD)	0.342	0.350	0.115	1.351	18
Volatility of exchange rate changes (StD)	0.014	0.006	0.006	0.030	18
Fama coefficient (UIP)	-0.493	0.789	-1.479	1.194	18
R-Squared (Exchange rate disconnect)	0.037	0.064	0.000	0.261	18

Notes: The table presents summary statistics for various moments of exchange rates across different currencies. Panel A. reports statistics calculated at a monthly frequency, while Panel B. reports statistics calculated at a weekly frequency. We calculated the moments for each currency individually and then compiled a table to summarize the variation of these moments across currencies. The data cover the period from 1993 to 2019 for 18 currencies as described in Appendix A.

Figure 5: Distribution Exchange Rate Expectations



Notes: The figure shows the distribution of the same-quarter EUR/USD exchange rate expectations from the ECB Professional Forecasters survey. Data covers the period from 2002Q1 to 2020Q4 and is collected at a quarterly frequency. Expectations are in log and demeaned at the quarterly frequency. Table 4 in Appendix A provides additional measures of the dispersion of exchange rate expectation across horizon and time periods.

Table 4: Summary Statistics - Expectation Dispersion

	Whole Sample	Average across Quarters	Median across Quarters
Same Quarter	0.028	0.024	0.020
Across all Horizons	0.041	0.038	0.035

Notes: The table reports the standard deviation of EUR/USD exchange rate expectations from the ECB Professional Forecasters survey. Data covers the period from 2002Q1 to 2020Q4 and is collected at a quarterly frequency for various horizons ranging from the same quarter to one year ahead. The expectations are expressed in logarithmic form to maintain consistency with the log-linearized model. Expectations are demeaned at the quarterly-horizon level. The first row focuses on same-quarter expectations, while the second row considers all horizons pooled together. The first column reports the dispersion (standard deviation) in exchange rate expectations across the whole sample period. The second and third columns compute the dispersion for each quarter and report the average and median dispersion across all quarters, respectively.

B Derivations

B.1 Derivation Demand Functions - Rational Expectation Case

Each investor j solves the following problem:

$$\begin{aligned} \max_{b_t^j} & \mathbb{E}_t^j(w_{t+1}^j | \Omega_t^j) - \frac{\rho}{2} \text{Var}_t^j(w_{t+1}^j | \Omega_t^j) \\ \text{s.t. } & w_{t+1}^j = (\omega - b_t^j)i_t + (i_t^* + s_{t+1} - s_t)b_t^j \end{aligned}$$

We assume that investors have symmetric rational expectation information sets, so that all j indexes on expectation and variance are dropped. We take the derivative of the objective function w.r.t. b_t^j . If the investor is strategic ($j = S$), they internalize the effect of their demand on the exchange rate. Thus, the demand schedule is:

$$b_t^{S,i} = \frac{\mathbb{E}_t(s_{t+1}) - s_t + i_t^* - i_t}{\rho \text{Var}_t(s_{t+1}) + \frac{\partial s_t}{\partial b_t^{S,i}}},$$

where the $\frac{\partial s_t}{\partial b_t^j}$ represents the price impact. If the investor is competitive ($j = C$), the demand schedule follows a standard mean-variance specification:

$$b_t^C = \frac{\mathbb{E}_t(s_{t+1}) - s_t + i_t^* - i_t}{\rho \text{Var}_t(s_{t+1})}.$$

We can now derive an expression for the price impact of a strategic investor. Assume there are N strategic investors, each with positive mass λ_i . Then, the market clearing condition for the foreign bond market is:

$$(1 - \lambda)b_t^C + \sum_i^N \lambda_i b_t^{S,i} + (x_t + \bar{x})\bar{W} = B(1 + s_t).$$

Substituting the demand schedule and applying the Implicit function theorem, we can write:

$$(1 - \lambda) \frac{\partial b_t^C}{\partial s_t} \frac{\partial s_t}{\partial b_t^{S,i}} + \lambda_i = B \frac{\partial s_t}{\partial b_t^{S,i}}$$

Thus:

$$\frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i}{B - (1 - \lambda) \frac{\partial b_t^C}{\partial s_t}} \quad \text{with} \quad \frac{\partial b_t^C}{\partial s_t} \equiv -\frac{1}{\rho \text{Var}_t(s_{t+1})}$$

Therefore:

$$\frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i \rho \text{Var}_t(s_{t+1})}{B \rho \text{Var}_t(s_{t+1}) + (1 - \lambda)} \equiv \frac{1}{N} \frac{\lambda \rho \sigma_t^2}{B \rho \sigma_t^2 + (1 - \lambda)} > 0$$

where the last equality holds in case of a symmetric oligopoly (i.e. $\lambda_i = \frac{\lambda}{N} \forall i$). The price impact is positive for $\forall (B, \lambda, N, \lambda_i, \rho, \sigma)$.

Lastly, in international portfolio choice models, the value of the supply of foreign assets in domestic currency (indirectly) depends on the value of the exchange rate when foreign assets are denominated in foreign currency. Differently from standard models of strategic trading (Kyle, 1989), strategic investors internalize not only their price effect on the quantity demanded but also on the quantity (value) supplied. Compared to closed economy models or cases in which foreign assets are denominated in domestic currency, the presence of this valuation effect on the supply implies a weakly lower price impact. Let pi^F and pi^D be the price impact on a foreign and a domestic asset, respectively.

$$pi^F \equiv \frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i \rho \sigma_t^2}{B \rho \sigma_t^2 + (1 - \lambda)} \quad pi^D \equiv \frac{\partial p_t}{\partial b_t^{S,i}} = \frac{\lambda_i \rho \sigma_t^2}{(1 - \lambda)}$$

where p_t is the price of the domestic asset. It is easy to show that $pi^F \leq pi^D \quad \forall (B, \rho, \sigma_t^2, \lambda_i, \lambda)$. The intuition is fairly simple. The increase in the price of a currency (foreign currency appreciates) increases the nominal value of the supply of foreign assets when denominated in domestic currency. The supply shift dampens the initial rise in price, reducing the magnitude of the price impact. The overall effect of tradings on the exchange rate is lower due to the presence of a valuation effect. In other words, the residual net demand faced by strategic investors is more elastic than in a case with no valuation effects. The main implication is that strategic investors still reduce their exposure to foreign assets compared to competitive investors but not as much as in the case there was no valuation effect.

B.2 Effect of Strategic Behavior on Noise and Fundamental Shock

The presence of strategic investors amplifies (dampens) the response of the exchange rate to noise (fundamental) shocks.

Proof. Consider the law motion of the exchange rate in Equation (7). s_t can be rewritten as a forward looking sum of fundamentals and noises as follow:

$$s_t = -\mu \sum_{k=0}^{\infty} \mu^k (\Delta i_{t+k}) + \frac{1-\mu}{b} \sum_{k=0}^{\infty} \mu^k (x_{t+k}),$$

where $\Delta i_{t+k} = i_{t+k} - i_{t+k}^*$. Therefore, the response of the exchange rate to a unit shock in noise and fundamental at impact is:

$$\mathbb{E}(s_{t+j}, j=0) = \begin{cases} \frac{\mu}{1-\mu\rho_u}, & \text{for } \varepsilon_u = -1 \\ \frac{(1-\mu)}{(1-\mu\rho_x)b}, & \text{for } \varepsilon_x = 1 \end{cases}$$

Taking the derivative w.r.t. μ , we find:

$$\frac{\partial \mathbb{E}(s_{t+j}, j=0)}{\partial \mu} = \begin{cases} \frac{1}{(1-\mu\rho_u)^2} > 0 \\ -\frac{(1-\rho_x)}{(1-\mu\rho_x)^2 b^2} < 0 \end{cases}$$

Since μ is decreasing (increasing) function of λ (N), the response of the exchange rate to a unit shock in fundamental is dampened while noise shock are amplified as λ increases (N decreases). ■

B.3 Excess Return Predictability

Strategic behavior amplifies these UIP deviations compared to a competitive market.

Proof. Define the one-period excess return as $q_{t+1} = s_{t+1} - s_t - (i_t - i_t^*)$. The one-period Fama regression is given by:

$$q_{t+1} = \alpha + \beta(i_t - i_t^*) + \epsilon_t$$

Now, consider the law of motion for the exchange rate, from Equation 7:

$$s_t = \mu [E_t(s_{t+1}) + i_t^* - i_t] + (1 - \mu) \frac{\bar{x}}{b} + (1 - \mu) \frac{1}{b} x_t,$$

where only the first term depends on fundamentals. By manipulating this equation, we can derive the j -period change in the exchange rate as follows:

$$\Delta s_{t+j} = -\mu \sum_{k=0}^{\infty} \mu^k (\Delta i_{t+j+k} - \Delta i_{t+k}).$$

Using Δs_{t+1} , we can then calculate the Fama coefficient is given by:

$$\begin{aligned} \beta_1 &= \frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t; \Delta i_t)}{\text{Var}(\Delta i_t)} = \left[\text{Cov} \left(-\mu \sum_{k=0}^{\infty} \mu^k (\Delta i_{t+k+1} - \Delta i_{t+k}); \Delta i_t \right) - \text{Var}(\Delta i_t) \right] / \text{Var}(\Delta i_t) \\ &= \left[-\mu \sum_{k=0}^{\infty} \mu^k \text{Cov}(\Delta i_{t+k+1} - \Delta i_{t+k}; \Delta i_t) - \text{Var}(\Delta i_t) \right] / \text{Var}(\Delta i_t) \\ &= \left[-\mu \sum_{k=0}^{\infty} \mu^k \rho_u^k (\rho_u - 1) \text{Var}(\Delta i_t) - \text{Var}(\Delta i_t) \right] / \text{Var}(\Delta i_t) \\ &= -(1 - \mu) \frac{1}{1 - \mu \rho_u} < 0, \end{aligned}$$

which is negative for each value of μ and increasing (decreasing) in μ (in λ). ■

Predictability J-Periods Ahead Predictability reversal does not arise in our model, differently from [Bacchetta and Van Wincoop \(2010\)](#) and [Engel \(2016\)](#). Formally define the j -period ahead excess return as $q_{t+j} = s_{t+j+1} - s_{t+j} - (i_{t+j} - i_{t+j}^*)$, and consider the following regression:

$$q_{t+j} = \alpha + \beta_j (i_t - i_t^*) + \epsilon_{t+j}. \quad (18)$$

The coefficient of interest, β_j , is:

$$\begin{aligned}
\beta_j &= \frac{\text{Cov}(q_{t+j}, \Delta i_t)}{\text{Var}(\Delta i_t)} \\
&= \frac{1}{\text{Var}(\Delta i_t)} (\text{Cov}(\Delta s_{t+j}, \Delta i_t) - \text{Cov}(\Delta i_{t+j-1}, \Delta i_t)) \\
&= \frac{1}{\text{Var}(\Delta i_t)} \left[\text{Cov} \left(-\mu \sum_{k=0}^{\infty} \mu^k (\Delta i_{t+k+j} - \Delta i_{t+k+j-1}); \Delta i_t \right) - \text{Cov}(\Delta i_{t+j-1}, \Delta i_t) \right] \\
&= \frac{1}{\text{Var}(\Delta i_t)} \left[\left(-\mu \sum_{k=0}^{\infty} \mu^k \text{Cov}(\Delta i_{t+k+j} - \Delta i_{t+k+j-1}; \Delta i_t) \right) - \text{Cov}(\Delta i_{t+j-1}, \Delta i_t) \right] \\
&= -\mu \sum_{k=0}^{\infty} \mu^k (\rho_u^{k+j} - \rho_u^{k+j-1}) - \rho_u^{j-1} \\
&= -\mu \rho_u^{j-1} (\rho_u - 1) \frac{1}{1 - \mu \rho_u} - \rho_u^{j-1} = -\rho_u^{j-1} \frac{1 - \mu}{1 - \mu \rho_u} \leq 0.
\end{aligned}$$

Lastly, notice that $\frac{\partial \beta_j}{\partial j} = -(j-1)\rho_u^{j-1} \left(\frac{1-\mu}{1-\mu\rho_u} \right) < 0$. Therefore, for $j \rightarrow \infty$, the coefficient $\beta_j \rightarrow 0$ monotonically, excluding any reversal.²⁰

B.4 Non-Monotonicity of Excess Volatility

The unconditional volatility of the exchange rate is non-monotonic in the presence of strategic investors.

Proof. Consider the law of motion of the exchange rate, Equation 7, and substitute the process for fundamental and noise:

$$s_t = -\mu \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \mu^k \rho_u^j \varepsilon_{t+k-j}^u + \frac{1-\mu}{b} \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \mu^k \rho_x^j \varepsilon_{t+k-j}^x.$$

After some algebra, s_t can be written as summation of its backward and forward components:

$$s_t = -\frac{\mu}{1 - \mu \rho_u} \left[\sum_{k=0}^{\infty} \mu^k \varepsilon_{t+k}^u + \sum_{k=1}^{\infty} \rho_u^k \varepsilon_{t-k}^u \right] + \frac{1 - \mu}{b(1 - \mu \rho_u)} \left[\sum_{k=0}^{\infty} \mu^k \varepsilon_{t+k}^x + \sum_{k=1}^{\infty} \rho_x^k \varepsilon_{t-k}^x \right].$$

²⁰This is not surprising considering the absence of any friction, such as infrequent portfolio adjustment (Bacchetta and Van Wincoop, 2010, 2019).

Thus, the unconditional variance of the exchange rate is:

$$\text{Var}(s) = \frac{\mu^2 \sigma_u^2}{(1 - \mu \rho_u)^2} \left[\frac{1}{1 - \mu^2} + \frac{\rho_u^2}{1 - \rho_u^2} \right] + \frac{(1 - \mu)^2 \sigma_x^2}{(1 - \mu \rho_x)^2 b^2} \left[\frac{1}{1 - \mu^2} + \frac{\rho_x^2}{1 - \rho_x^2} \right],$$

which is a combination of the variances of fundamental and noise shocks. Taking the derivative of $\text{Var}(s)$ w.r.t. μ , we find:

$$\begin{aligned} \frac{\partial \text{Var}(s)}{\partial \mu} = & \frac{\mu \sigma_u^2}{(1 - \mu \rho_u)^3} \left[\frac{1}{1 - \mu^2} + \frac{\rho_u^2}{1 - \rho_u^2} \right] + \frac{\mu^3 \sigma_u^2}{(1 - \mu \rho_u)^2 (1 - \mu^2)^2} - \\ & \frac{(1 - \mu)(1 - \rho_x) \sigma_x^2}{(1 - \mu \rho_x)^3 b^2} \left[\frac{1}{1 - \mu^2} + \frac{\rho_x^2}{1 - \rho_x^2} \right] + \frac{\mu(1 - \mu)^2 \sigma_x^2}{(1 - \mu \rho_x)^2 (1 - \mu^2)^2 b^2}. \end{aligned}$$

The unconditional volatility of the exchange rate is increasing in λ iff:

$$\begin{aligned} & \frac{(1 + \mu \rho_x) \sigma_x^2}{(1 - \mu \rho_x)^2 (1 + \mu)(1 + \rho_x) b^2} - \frac{\mu \sigma_x^2}{(1 - \mu \rho_x)^2 (1 + \mu)^2 b^2} > \\ & \frac{\mu \sigma_u^2}{(1 - \mu \rho_u)^2} \frac{(1 + \mu \rho_u)}{(1 - \mu^2)(1 - \rho_u^2)} + \frac{\mu^3 \sigma_u^2}{(1 - \mu \rho_u)^2 (1 - \mu^2)^2}, \end{aligned}$$

that can be rewritten as follows:

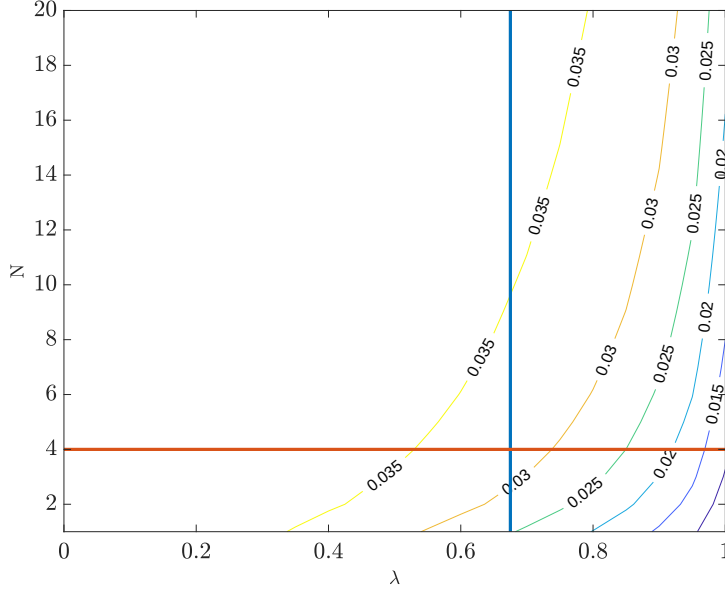
$$\frac{\text{Var}(x)}{\text{Var}(\Delta i)} \frac{1}{b^2} > \left[\frac{(1 + \mu^2 \rho_x)(1 - \rho_x)}{\mu(1 + \mu \rho_u)(1 - \mu^2) + \mu^3(1 - \rho_u^2)} \frac{(1 - \mu \rho_u)^2 (1 - \mu)^2}{(1 - \mu \rho_x)^2} \right]^{-1}. \quad (19)$$

■

Equation (19) suggests that the unconditional variance of the exchange rate increases as λ increases when the variance of the noise shock is sufficiently high compared to the variance of the fundamental process.

The non monotonic case is not relevant given standard parametrizations, including ours. Let define $\underline{\sigma}_x$ as the minimum value of the volatility of the noise process at which the relationship between the level of strategic behavior and exchange rate variance becomes non-monotonic. Figure 6 shows the value of $\underline{\sigma}_x$ for different combinations of N and λ . In our calibration, we find that the volatility of the noise shock should be at least 75% lower in order to break the monotonic relationship between strategic behavior (λ and/or N) and the unconditional variance of the exchange rate. In cases where λ or N take on other values, the minimum value of σ_x is at least 50% lower compared to the value implied by Figure 8 in Appendix D. For instance, in a market with a high level of strategic behavior

Figure 6: $\underline{\sigma}_x$ for different combinations of N and λ .



Notes: The figure shows the minimum value of the volatility of the noise process, σ_x , that guarantees that the volatility of the exchange rate is monotonically increasing in the presence of strategic behavior (higher λ and/or lower N). The threshold is computed using Equation (19). We compute the minimum value of σ_x for different levels of λ and N . The horizontal and vertical lines pin down the combination of λ and N used in the parametrization of the basic framework. Remaining parameters are constant, see Table 6.

(λ approximately 1), we find that σ_x is approximately 0.05. However, monotonicity in the relationship between strategic behavior and unconditional variance breaks if σ_x falls below 0.025.

Furthermore, it is important to note that the threshold value mentioned earlier is dependent on the parameters ρ_x , ρ_u and b . The robustness of the monotonic relationship between strategic behavior and unconditional variance is also guaranteed by the conservative nature of our calibration. In standard calibrations, only more persistent noise processes or less persistent fundamental processes would align with the observed data. Similarly, higher values of home bias (lower b) would be consistent with the data. Higher values of ρ_x , lower values of ρ_u and lower b all contribute to reducing the threshold, thereby relaxing the condition for monotonicity.

B.5 Exchange Rate Disconnect

The effect of strategic investors on exchange rate disconnect is ambiguous.

Proof. Define the exchange rate change as $\Delta s_{t+1} = s_{t+1} - s_t$ and the interest rate differential as Δi_t . The exchange rate disconnect can be measured by the R^2 of the following regression:

$$\Delta s_{t+1} = \alpha + \beta \Delta i_t + \varepsilon_t$$

The R^2 is given by:

$$R^2 = \text{Corr}(\Delta s_{t+1}, \Delta i_t)^2 = \frac{\text{Cov}(\Delta s_{t+1}, \Delta i_t)^2}{\text{Var}(\Delta s_{t+1}) \text{Var}(\Delta i_t)} = \left[\frac{\text{Cov}(\Delta s_{t+1}, \Delta i_t)}{\text{Var}(\Delta i_t)} \right] \left[\frac{\text{Cov}(\Delta s_{t+1}, \Delta i_t)}{\text{Var}(\Delta s_{t+1})} \right].$$

Adding and subtracting $\text{Var}(\Delta i_t)$ in the numerator, we have:

$$\begin{aligned} R^2 &= \left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t)}{\text{Var}(\Delta i_t)} + 1 \right] \left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t) + \text{Var}(\Delta i_t)}{\text{Var}(\Delta s_{t+1})} \right], \\ &= \left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t)}{\text{Var}(\Delta i_t)} + 1 \right] \frac{\text{Var}(\Delta i_t)}{\text{Var}(\Delta s_{t+1})} \left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t)}{\text{Var}(\Delta i_t)} + 1 \right]. \end{aligned}$$

Since the covariance term is the Fama coefficient β^{Fama} , it follows that:

$$R^2 = \frac{\text{Var}(\Delta i_t)}{\text{Var}(\Delta s_{t+1})} [1 + \beta^{Fama}]^2.$$

■

B.6 Cross-Currency Testable Model Predictions

Table 5: Testing Model Predictions

	(1)	(2)	(3)
	Fama Coefficient	Excess Volatility	Disconnect - R ²
λ	-43.759*** (8.288)	180.925*** (69.343)	-0.248*** (0.068)
Constant	29.304*** (4.684)	53.229 (38.980)	0.232*** (0.038)
Currency & Year FEs	Yes	Yes	Yes
Observations	900	900	900

Notes: The table reports the relationship between λ and the variables of interest. λ represents the net concentration ratio by the top eight reporting traders operating in the future FX market. Variable of interest are: Fama coefficient (Column (1)), exchange rate excess volatility (Columns (2)); exchange rate disconnect/R² (Column (3)). The exchange rate disconnect is measured using the Adjusted R² from the regression in Equation (16), while excess volatility is calculated as the ratio of exchange rate volatility from Equation (14) to the volatility of the interest rate differential. λ is measured monthly from 2006 to 2016 using the U.S. CFTC data. To measure excess volatility and disconnect, we use a two-year periods rolling window and exchange rate data at monthly level. Values of the excess volatility ratio and disconnect are winsorized at 1%. All regressions include currency and year fixed effects. Standard errors in parenthesis are clustered at the currency level. Significance level: * p<0.10, ** p<0.05, *** p<0.01. Currencies considered are: Euro, Japanese Yen, Brazilian Real, Canadian Dollar, Swiss Franc, Australian Dollar, Mexican Peso, British Pound, Russian Ruble, and New Zeland Dollar. Appendix A provides additional information on the data used.

C Quantitative Assessment

C.1 A Benchmark Calibration

Table 6: Benchmark Parametrization

Parameters	Value	Target
ρ	50	Bacchetta and Van Wincoop (2019)
b	0.333	Home Bias
ρ_u	0.85	Average persistence AR(1) Δi_t
σ_u	0.005	Average StD innovation AR(1) Δi_t
σ_t	0.028	σ_t (Volatility ER change)
ρ_x	0.9	ER Random Walk/Average Disconnect
λ	0.572	Net concentration ratio (Top 8) – US CFTC
N	8	Number of traders related to λ – US CFTC
σ_x	0.132	Average StD ER change

Notes: The table summarizes the parametrization used in the basic framework. For each parameters, we report the value used in the model, the corresponding moment and data used to calibrate, and, if applicable, the target moment used to estimate it. [Appendix A](#) provides additional information on the data used.

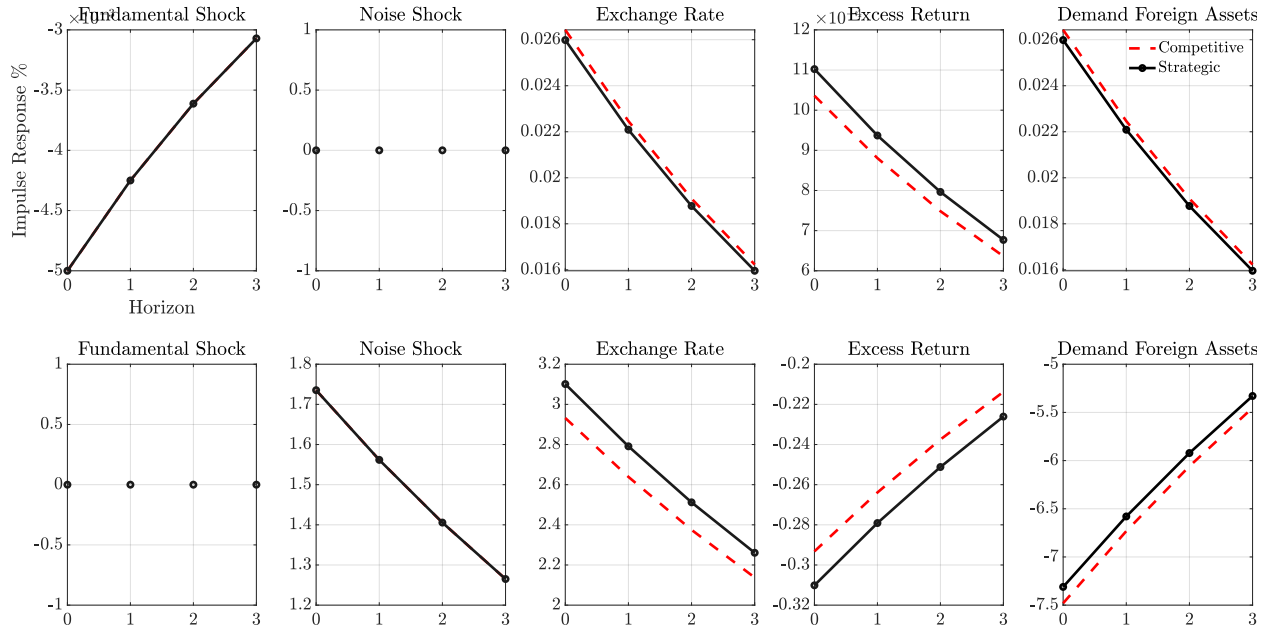
C.2 Impact of Strategic Investors on Exchange Rates

Table 7: Potential Impact of Strategic Investors in Benchmark Model

	(1)	(2)	(3)	(4)
Panel A: Moments from Simulated Exchange Rate Data	Response to Shock Noise (Fundamental)	Fama Coefficient	Exchange Rate Volatility (St.D.)	Exchange Rate Disconnect
Competitive (Atomistic investors)	14.66 (-0.088)	-0.204	0.259	0.083
Strategic (Avg. investor 7%)	15.506 (-0.087)	-0.217	0.274	0.073
Periods per Simulation	6000	6000	6000	6000
Burn-in Periods	3000	3000	3000	3000
Number of Simulations	1000	1000	1000	1000
Panel B: Moments from Actual Exchange Rate Data		Fama Coefficient	Exchange Rate Volatility (St.D.)	Exchange Rate Disconnect
Monthly Avg. Across Currencies (St.D.)		-0.603 (1.317)	0.344 (0.353)	0.030 (0.065)
Weekly Avg. Across Currencies (St.D.)		-0.490 (0.780)	0.341 (0.339)	0.036 (0.064)
Years per Currency		1993-2019	1993-2019	1993-2019
Number of Currencies		18	18	18

Notes: Panel A reports the impact of strategic investors on exchange rate dynamics using simulated data, with results for the competitive and benchmark models in the first and second rows, respectively. Column (1) shows the exchange rate response to a one-standard-deviation noise and fundamental shock, expressed as a percentage deviation from the steady state (e.g., 1 indicates a 1% deviation). Column (2) displays the Fama regression beta, indicating deviations from uncovered interest parity. Column (3) reports exchange rate volatility (standard deviation), and Column (4) provides the estimated R^2 from the disconnect regression in the simulated data. Panel B shows the average and standard deviation of equivalent moments from actual exchange rate data at monthly and weekly frequencies. The results in Panel A are based on 1,000 simulations, each running for 6,000 periods with a 3,000-period burn-in. All other parameters are constant across scenarios; see Table 6 for details. Appendix A provides further data information, and Appendix C outlines the estimation procedure.

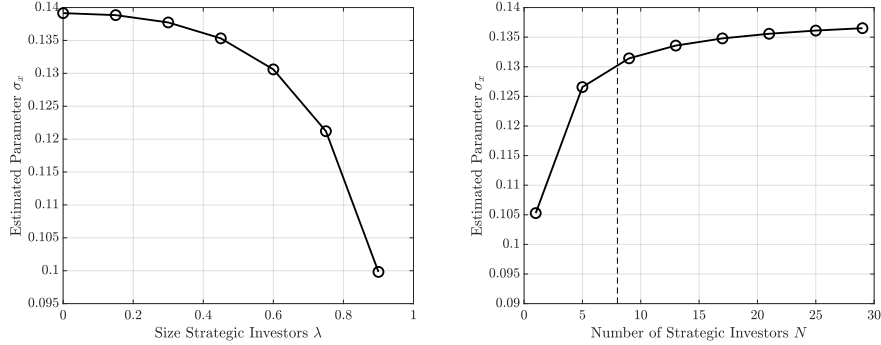
Figure 7: Impulse Response to Exogenous Shocks



Notes: The top panel (bottom) shows the response to a two-standard deviation shock in fundamental and noise. The first and second columns show the dynamics of the exogenous shocks in fundamentals and noise, respectively. The third column shows the dynamics of the exchange rate. Column four shows the response of the realized excess return, defined as $q_{t+1} = s_{t+1} - s_t + i_t^* - i_t$. The last column shows the response of the total demand of foreign assets, defined as $(1 - \lambda)b_t^C + \sum_i^N \lambda_i b_t^{S,i}$, where b_t^C and $b_t^{S,i}$ are defined according to Equation 4. The solid black line shows the response in the benchmark parametrization with strategic investors, $\lambda = 0.6$. The red dashed line shows the response in a competitive economy without strategic investors, $\lambda = 0$. Remaining parameters are common across scenarios, see Table 6.

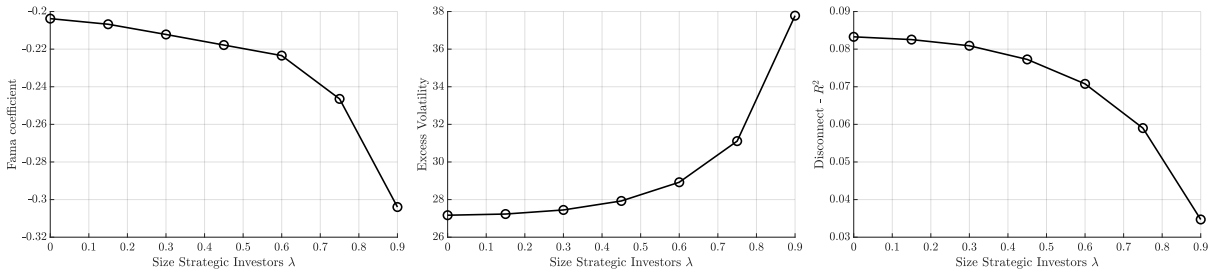
C.4 Potential Impact of Strategic Investors

Figure 8: Relationship between Strategic Behavior and Noise Volatility



Notes: The figure shows the volatility of the noise component, σ_x , required to match the target volatility of the exchange rate change in the basic framework, for different levels of strategic behavior. The left panel considers different levels of strategic behavior in terms of λ for a number of strategic investors equal to $N = 4$. The right panel considers different levels of strategic behavior in terms of N for a total size of strategic investors equal to $\lambda = 0.675$. All other parameters are constant and summarized in Table 6.

Figure 9: Exchange Rate Disconnect and Excess Volatility



Notes: The figure shows the excess predictability coefficient from regression in Equation 11 (top left), the excess volatility ratio (top right), and the estimated R^2 of the disconnect regression in Equation 16 (bottom) using simulated data. We run 5000 simulations and, for each iteration, the model runs for 8000 periods with 3000 burn-in. Data are simulated for different levels of strategic behavior λ . Remaining parameters are common across scenarios, see Table 6.

D Extending the Model with Dispersed Information

We now compare the effects of heterogeneity in price impact have on excess volatility and disconnect to the effects of another well-established dimension of heterogeneity: investors' information heterogeneity. Dispersed information arising from heterogeneous information sets may leads to higher exchange rate disconnect and excess volatility (Bacchetta and Van Wincoop, 2006; Evans and Lyons, 2002), representing a competing mechanism with heterogeneity in price impact. To assess the relevance of these two competing dimensions of heterogeneity, we extend the basic framework presented in Section 2 by relaxing the full information assumption and including information heterogeneity based on Nimark (2017). Through the lens of our model, we quantitatively evaluate the relative importance of strategic behavior and information heterogeneity in driving the dynamics of exchange rates.

D.1 Model

The model incorporates all standard elements of an exchange rate monetary model, along with the strategic behavior described in Section 2. However, in contrast to the basic framework, we assume that investors possess imperfect knowledge of the shocks affecting the economy, resulting in dispersed information. The remaining structure of the economy remains the same.

The main implication of information heterogeneity is that the optimal demand for foreign bonds by investor j at time t now depends on their individual information set, $\Omega_t(j)$:

$$b_t^j = \begin{cases} \frac{\mathbb{E}_t(s_{t+1}|\Omega_t(j)) - s_t + i_t^* - i_t}{\rho\sigma_t^2} & \text{if } j = C \\ \frac{\mathbb{E}_t(s_{t+1}|\Omega_t(j)) - s_t + i_t^* - i_t}{\rho\sigma_t^2 + \frac{\partial s_t}{\partial b_t^j}} & \text{if } j = S \end{cases} \quad (20)$$

where the excess return, $q_{t+1} = E_t(s_{t+1}|\Omega_t(j)) - s_t + i_t^* - i_t$, and the variance of the exchange rate change, σ_t^2 , are now conditional to the information set at time t , $\Omega_t(j)$. In contrast to the basic framework, we assume that σ_t^2 is endogenous but common to all investors, implicitly assuming that investors have the same capacity to process information. Despite the presence of information heterogeneity, the main implication of strategic behavior still holds true. Specifically, the own price impact reduces the demand of strategic investors

for any given level of excess return.

Information Structure The information structure in our model follows [Nimark \(2017\)](#), and generalize the case in [Singleton \(1987\)](#) and [Bacchetta and Van Wincoop \(2006\)](#). Investors form expectation regarding the future price of the foreign bond (exchange rate) by observing their private signal about the fundamental, as well as the history of the exchange rate. Formally, investors' information set is given by:

$$\Omega_t(j) = \{f_{t-T}(j), s_{t-T} : T \geq 0\},$$

where

$$f_t(j) = \Delta i_t + \eta_t(j) \text{ where } \eta_t(j) \sim N(0, \sigma_\eta^2)$$

represents the private signal about fundamentals. Therefore, investors have imperfect knowledge about the history of shocks that affect the economy because they observe an unbiased signal $f_t(j)$ regarding Δi_t , with an idiosyncratic measurement error $\eta_t(j)$. Investors are unable to perfectly observe the path of the foreign interest rate, and cannot deduce the fundamental component from observing the exchange rate due to the presence of unobserved transitory noise shock x_t ([Admati, 1985](#)). The private signal, $\eta_t(j)$, implies that investors have different expectations about foreign Central Bank's operating procedures. Consequently, the need to 'forecast the forecasts of others' (infinite regress problem) arises due to information dispersion.

The key distinction with [Singleton \(1987\)](#) and [Bacchetta and Van Wincoop \(2006\)](#) lies in the nature of private signals, which are not short-lived. In other words, innovations to the fundamental process are not perfectly and publicly observed after a finite number of periods. Short-lived private information allows to derive a finite dimensional state representation, overcoming the infinite regress problem. The solution method proposed by [Nimark \(2017\)](#) and used here enables us to solve our model while relaxing the assumption made by [Singleton \(1987\)](#).

Equilibrium We extend the definition of equilibrium of the basic framework discussed in [Section 2](#) to incorporate the presence of dispersed information. In the extended framework, an equilibrium path is defined as a sequence of quantities $\{b_t(j)\}$ and foreign currency (asset) price $\{s_t\}$ that satisfy the following conditions: given an history of shocks $\{\varepsilon_t^x\}_{t=0}^{-\infty}$ and signals about fundamentals $\{f_t(j)\}_{t=0}^{-\infty}$, investors optimally choose their portfolios, and the market

clearing condition is upheld.

The effect of strategic behavior on the exchange rate, as well as its mechanism, extends to the model with dispersed information as in the basic framework. Combining the market clearing condition with investors' demand schedules, we can derive the following expression for the exchange rate:

$$s_t = (1 - \mu) \left(\frac{\bar{x}}{b} - 1 \right) + \mu \left(\int \mathbb{E}[s_{t+1} | \Omega_t(j)] dj \right) - \mu (i_t - i_t^*) + (1 - \mu) \frac{1}{b} x_t, \quad (21)$$

where μ and Φ are defined as in the basic framework, with the former decreasing in the presence of strategic investors (decreasing in λ and increasing in N).

In the presence of dispersed information, a closed-form solution for the exchange rate is not available since it depends on higher-order expectations regarding the fundamental:

$$s_t = \mu \sum_{k=0}^{\infty} \mu^k [i_{t+k} - i_{t+k}^*]_t^{(k)} + \frac{1 - \mu}{b} x_t, \quad (22)$$

where $[i_{t+k} - i_{t+k}^*]_t^{(k)}$ denotes the average expectation in period t of the average expectation in period $t+1$, and so on, of the average expectation in period $t+k-1$ of k period ahead fundamentals, that is, $[i_{t+k} - i_{t+k}^*]_t^{(k)} = \underbrace{\int \mathbb{E}_t \dots \left[\int \mathbb{E}_{t+k-1} (i_{t+k} - i_{t+k}^*) dj \right] \dots dj}_k$ for any integer

$k > 0$. In the case of dispersed information, the informativeness parameter μ represents the weight assigned to higher-order expectations regarding future fundamentals in influencing exchange rate dynamics.

To account for higher order expectations, we assume that agents have rational expectations about how other agents form their own expectations, and that this information is common knowledge. Using this assumption, we compute the dynamics of the exchange rate while accounting for expectations of arbitrarily high orders. Denoting the hierarchy of expectations about fundamentals with $\Delta i_t^{(0:k)}$, which is the vector of average expectations on Δi_t of any order from zero to k , we show below that the exchange rate s_t can be expressed as:²¹

$$s_t = v_0 \Delta i_t^{(0:k)} + \frac{1 - \mu}{b} x_t \quad (23)$$

²¹There exist other approaches that rely on the fact that average first-order expectations about the endogenous variables can be computed given the guessed laws of motion of the endogenous variables by using the assumption of rational expectations. We find the approach in [Nimark \(2017\)](#) more reliable and fast to implement.

where v_0 is a vector of k weights associated to higher order expectations. In contrast to the baseline model, an aggregate shock in this model affects the exchange rate not only directly, but also through higher order expectations $\Delta i_t^{(1:k)}$.

D.2 Solution Method

We solve the model with higher order expectations using the recursive solution algorithm in [Nimark \(2017\)](#). We approximate the equilibrium of the model to an arbitrary precision with finite number of higher order expectations $\bar{k} < \infty$.

We recursively computes the exchange rate process and the law of motion of the expectations hierarchy for arbitrarily high orders of expectations following these steps:

Step 1. Define the zero order process ($k = 0$) for the exchange rate s_t as a function of the current fundamentals $\Delta i_t^{(0)}$:

$$\begin{aligned} s_t &= G_k \Delta i_t^{(0)} + R_1 \mathbf{w}_t \\ \Delta i_t^{(0)} &= M_k \Delta i_{t-1}^{(0)} + N_k \mathbf{w}_t \end{aligned}$$

where $\mathbf{w}_t$ is the vector of aggregate shocks, including both fundamental and noise shocks; R_1 and N_k represent the variance matrices associated with the zero-order state space representation; the matrix $G_k \equiv G_0 = -\mu$, and $M_k \equiv M_0 = \rho$ are stored separately in the zero-iteration period.

Because investors learn from the exchange rate s_t , the measurement equation for investor j at time t includes a noisy signal about Δi_t as well as s_t :

$$\mathbf{s}_{j,t} = D_0 \Delta i_t^{(0:k)} + R_1 \mathbf{w}_t + R_2 w_{j,t} \quad w_{j,t} \sim N(0, I)$$

where $D_0 = [1, G_0]'$ and $w_{j,t}$ is the idiosyncratic noise shock.

Step 2. Using the measurement equation and the law of motion of hierarchy, compute the Kalman gain K_k for the k^{th} step, as well as the matrices M_{k+1} and N_{k+1} :

$$M_{k+1} = \begin{bmatrix} M_0 & \mathbf{0}_{q \times kq} \\ \mathbf{0}_{kq \times q} & \mathbf{0}_{kq \times kq} \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{q \times kq} & \mathbf{0}_{q \times q} \\ K_k D_k M_k & \mathbf{0}_{kq \times q} \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{q \times q} & \mathbf{0}_{q \times kq} \\ \mathbf{0}_{kq \times q} & (I - K_k D_k) M_k \end{bmatrix}$$

$$N_{k+1} = \begin{bmatrix} N_0 \\ (K_k D_k N_k + K_k R_1) \end{bmatrix}.$$

to get the k^{th} step law of motion

$$\Delta i_t^{(0:k)} = M_{k+1} \Delta i_{t-1}^{(0:k)} + N_{k+1} \mathbf{w}_t, \quad \mathbf{w}_t \sim N(0, I)$$

where the matrix D_k is defined as:

$$D_k = \begin{bmatrix} 1 & \mathbf{0}_{q \times kq} \\ G_k & \end{bmatrix}$$

Step 3. The k -order process for the exchange rate s_t^{k+1} is:

$$s_t^{k+1} = G_{k+1} \Delta i_t^{(0:k+1)} + R_1 w_t$$

where

$$G_{k+1} = G_0 + \mu G_k M_k H_{k+1} \quad \text{and} \quad H_k \equiv \begin{bmatrix} \mathbf{0}_{(kq) \times q} & I_{kq} \end{bmatrix}$$

Step 4. Repeat Steps 2 – 3 for $k = 1, 2, \dots, \bar{k}$ where the number of iterations $\bar{k}$ can be chosen to achieve any desired degree of accuracy.

D.3 Mechanisms and Results

Dispersed Information and Exchange Rate Dynamics Similarly to the presence of strategic behavior in our basic framework, the presence of dispersed information also amplifies the effects of noise shocks on the exchange rate while dampening the effects of fundamental shocks. Information heterogeneity leads to rational confusion, which means that investors always revise their expectations whenever the exchange rate changes, independently of the underlying shock. This confusion arises because investors are uncertain whether the fluctuations in the exchange rate are driven by noise shocks or fundamental shocks. It follows that, after a negative fundamental shock, investors' expectation do not fully react because part of the response of exchange rates is attributed to the noise component. As a result, the response of exchange rate to a fundamental shock is dampened. Similarly, the response to a positive noise shock is amplified because the upward movements in the exchange rate are

mistakenly confused with a negative change in fundamentals. This rational confusion adds further upward pressure on the exchange rate.²² This indicates that these two dimension of heterogeneity have similar qualitative implications for the dynamics of the exchange rate, albeit through different mechanisms. Strategic behavior reduces the sensitivity of investors' demand for foreign assets, while information heterogeneity leads to rational confusion.

Parametrization We extend the parametrization of the basic framework in Table 6 to account for the presence of dispersed information. We leverage the data on exchange rate expectations from the ECB Professional Forecasters survey to calibrate the volatility of the private signal, σ_η . The survey runs at quarterly frequency since 2002 and contains information on professional forecasters' expectations for the euro-dollar exchange rate at various horizons. Figure 5 reports the distribution of the demeaned, same-quarter exchange rate expectations. It exhibits a significant dispersion, with a standard deviation of approximately 0.02, indicating the presence of information heterogeneity among investors.²³ We estimate the volatility of the exchange rate change and the median dispersion in the same-quarter exchange rate forecasts across quarters using simulated methods of moments. This yields $\sigma_x = 0.024$ and $\sigma_\eta = 0.006$.

Quantitative Analysis We utilize the model that incorporates the two dimensions of heterogeneity to assess which dimension of heterogeneity is potentially relatively more important. We assume that the model embedding both strategic behavior and dispersed information represents the actual data, and decompose the contributions of both elements to the dynamics of the exchange rate. We filter the underlying states and conduct three different counterfactual scenarios: i) a competitive, full-information rational expectation benchmark economy without strategic investors and dispersed information ($\lambda = \sigma_\eta = 0$); ii) an economy where investors have dispersed information but are not strategic ($\lambda = 0$ and $\sigma_\eta > 0$); iii) an economy where investors are strategic and have full-information ($\lambda > 0$ and $\sigma_\eta = 0$). We perform the decomposition for different initial level of strategic behavior ($\lambda \in \{0, 0.2, 0.4, 0.6, 0.8\}$), given the measurement noise in our proxy for strategic behavior. We focus on the exchange rate disconnect, which is measured by the RMSE of the

²²As standard in this class of models, the model produces endogenous persistence due to the time it takes for rational confusion to be resolved. This means that average and higher-order expectations gradually converge to the rational expectation benchmark based on full information over time.

²³Table 4 in Appendix A provides additional measures of the dispersion of exchange rate expectation across horizon and time periods.

Table 8: Disconnect and Volatility Decomposition

Mass Strategic Investors (%)	Extra Disconnect (%)	% Share Strategic Behavior	% Share Dispersed Information	Non linearity
0.00	16.76	0.00	100.00	0.00
20.00	17.18	2.97	96.88	0.15
40.00	18.91	13.74	85.59	0.67
60.00	23.57	34.65	63.79	1.56
80.00	37.67	65.42	32.14	2.44
Mass Strategic Investors (%)	Extra Volatility (%)	% Share Strategic Behavior	% Share Dispersed Information	Non linearity
0.00	5.18	0.00	100.00	0.00
20.00	5.66	8.90	91.00	0.11
40.00	7.62	33.70	65.92	0.39
60.00	12.87	62.81	36.57	0.62
80.00	28.61	85.72	13.74	0.53

Notes: The table reports the contribution of strategic behavior and dispersed information to the exchange rate disconnect (top panel) and the excess volatility (bottom panel) for different value of λ (first column). Exchange rate disconnect is measured using the RMSE of a standard, one-period disconnect pooled regression, Equation (16). Excess volatility is measured using the standard deviation of the exchange rate. The second column reports the extra disconnect and volatility of the full model relative to a benchmark economy that abstract away from both dispersed information and strategic behavior ($\lambda = 0$ and $\sigma_\eta = 0$). The third and fourth columns report the share of the extra disconnect and volatility due to dispersed information and strategic behavior, respectively. The former (latter) is computed comparing RMSE/volatility in the benchmark economy to the RMSE/volatility from an economy without strategic behavior, $\lambda = 0$ and $\sigma_\eta > 0$ (without dispersed information, $\lambda > 0$ and $\sigma_\eta = 0$). The last column reports the discrepancy due to the non-linear interaction between dispersed information and strategic behavior. We exclude the Argentinian Peso from calculation. Appendix A provides additional information on the data.

disconnect regression in Equation (16), and the exchange rate excess volatility, which is measured by the volatility of the exchange rate.

Table 8 shows that investors' heterogeneity can have significant impact on exchange rate dynamics, and the relative importance of each dimension of heterogeneity greatly depends on the degree of strategic behavior in the market. Investors' heterogeneity increases exchange rate disconnect by 16% to 38% and volatility by 6% to 29%, playing a quantitatively significant role in shaping exchange rate dynamics, as highlighted in previous studies (Evans and Lyons, 2002; Bacchetta and Van Wincoop, 2006, 2010, 2019).²⁴

By comparing the competitive rational expectation model to an economy with only one

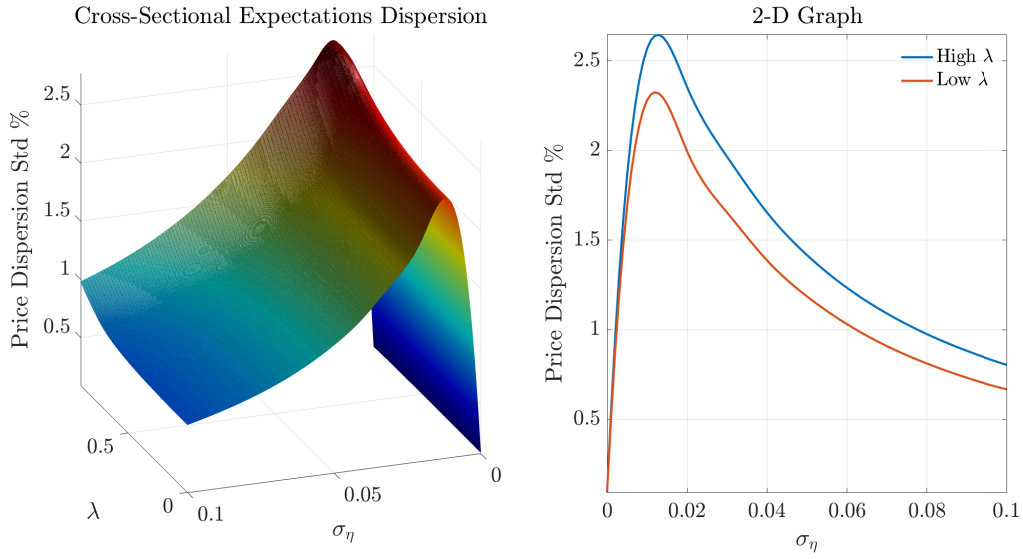
²⁴The economic relevance of the contribution of investors' heterogeneity extends beyond the changes in predictive power or in volatility, which may be relatively small in absolute terms. By influencing exchange rate dynamics, investors' heterogeneity has far-reaching implications for carry trade return, invoicing choices, relative international prices, trade patterns, and other aggregate macro variables (Boz et al., 2020; Itskhoki and Mukhin, 2021; Lustig et al., 2019).

dimension of heterogeneity, we show that the specific contributions of each individual dimension to exchange rate dynamics depends on the degree of strategic behavior. As λ increases, the contribution of strategic behavior rises from 3% to 66% for disconnect and from 9% to 86% for excess volatility. It's worth noting that the marginal effect of λ on the additional disconnect is lower than on additional volatility, suggesting that dispersed information appears to be more relevant in explaining exchange rate disconnect, regardless of the size of strategic investors in the market. Meanwhile, heterogeneity in price impact has a relatively more pronounced effect on excess volatility dynamics. These results highlight the quantitative relevance of both dimensions of investors' heterogeneity and underscore the importance of considering them in the analysis of exchange rate markets.

The final column in Table 8 shows that the response of the exchange rate in a model that incorporates both dispersed information and strategic behavior is not simply the sum of the individual mechanisms. Instead, there is a non-linear interaction between the two. For each value of λ , this non-linear interaction, accounting for approximately 0.1% to 2.5% of the overall effect, demonstrates that the two mechanisms reinforce each other. The idea is that strategic behavior leads to greater price dispersion regardless of the quality of the signal, σ_η . This, in turn, reduces the weight that investors assign to their signals and amplifies the impact of noise shocks while dampening the impact of fundamental shocks.²⁵

²⁵Figure 10 show the simulated price dispersion for different levels of strategic behavior and signal quality. Note that when the quality of the signal is sufficiently low (high σ_η), the volatility of the exchange rate may no longer increase. As the signal quality deteriorates, less importance is given to the fundamental component. This leads to a situation where the exchange rate becomes less informative, resulting in a reduction in the amplification of the noise component (Bacchetta and Van Wincoop, 2006).

Figure 10: Exchange Rate Expectation - Dispersion



Notes: The figure shows the dispersion (standard deviation) across investors in the one-period exchange rate expectations for different level of strategic behavior (λ) and precision of the signal on fundamentals (σ_η) implied by the model in Section ???. The left panel shows the dispersion in expectations for values of $\lambda \in [0, 1]$, and $\sigma_\eta \in [0, 0.1]$. The right panel shows the dispersion in expectation for two levels of strategic behavior ("Low" with $\lambda = 0$, and "High" with $\lambda = 0.6$) and a precision of the signal σ_η between 0 and 0.1. The figure is generated for a representative calibration with $\sigma_u = 0.01$ and $\rho_x = 0$.